

Fuzzy-Gated Verification of LLM Agents for Industrial Fault Diagnosis: An Evolutionary-Optimized Closed-Loop Framework

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Abstract

Intelligent fault diagnosis and machine health management are essential for reliable industrial operation, yet large language model (LLM) agents introduce reliability risks: hallucinated evidence, overconfident outputs, and inconsistent reasoning. We propose CL-LLMOps, a closed-loop LLMOps framework that routes LLM agents through five stages, from alarm normalization to regression-tested action. Its semantic verification layer combines rule-based pattern checking, cross-attention evidence scoring, and Mamdani fuzzy inference gating, jointly tuned by multi-objective particle swarm optimization. To our knowledge, soft computing has not been applied at this layer before. We evaluate CL-LLMOps on a synthetic power quality benchmark and two public datasets for bearing fault diagnosis and wind turbine SCADA anomalies, with open-source Llama-3.1-8B and Mistral-7B backbones. Under a paired protocol in which both arms share alarms, hypotheses, and evidence, verification cuts the released-hallucination rate from 43.7% to 17.7% on power quality data. On CWRU the rate falls from 24.5% to 1.5% and on CARE from 50.9% to 10.0%; all three reductions are significant under Holm-corrected Wilcoxon tests. The reductions hold under both hallucination judges, with sizes that depend on the judge, and survive resampling the verifier’s training set. Generic verification machinery does not reach this trade-off: a keyword guardrail leaves hallucination near the unverified rate, while an off-the-shelf entailment gate suppresses it only by escalating every episode. Evolutionary optimization exposes the hallucination–handoff trade-off as a Pareto front, and the deployed knee buys 10.7% hallucination for 58.9% human review on validation data. The framework bridges LLM capability and operational dependability for Industry 5.0 fault diagnosis by integrating neural attention, fuzzy logic, and evolutionary optimization at the verification layer.

Keywords: Industrial Fault Diagnosis, Machine Health Management, Large Language Model Agents, Hallucination Detection, Fuzzy Inference Gating, Evolutionary Optimization, Industry 5.0

1. Introduction

Intelligent fault diagnosis is a cornerstone of industrial asset management and machine health monitoring. It enables timely detection and remediation of equipment anomalies before they escalate into costly failures or safety hazards [1]. Traditional approaches based on signal processing and classical machine learning, such as support vector machines [2] and random forests, have demonstrated effectiveness when fault signatures are well-characterized and operating conditions remain stationary. However, these methods require labor-intensive feature engineering, struggle with distribution shifts, and produce only classification labels without actionable maintenance guidance [2]. Deep learning architectures, including convolutional neural networks [3] and Transformers [4, 5], have substantially improved diagnostic accuracy by learning discriminative representations from raw signals. Yet these models remain confined to classification and regression tasks. They cannot generate natural-language explanations, reason about evidence provenance, or produce structured action plans [6, 7]. The emergence of large language models (LLMs) offers a compelling path toward closing this capability gap.

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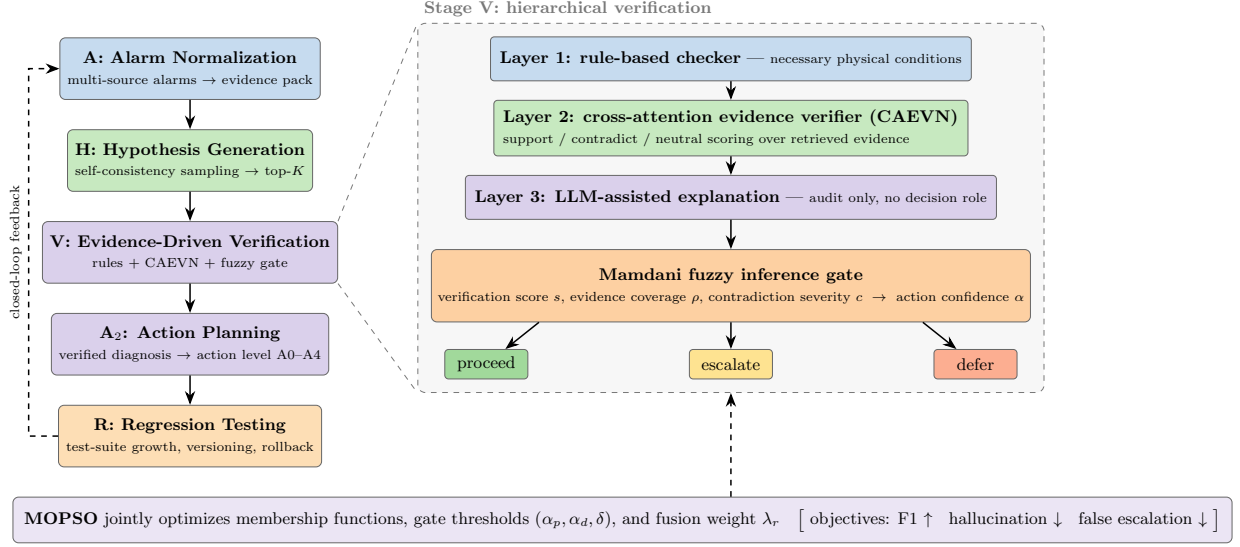


Figure 1: Overview of the CL-LLMOps framework. Five stages form a closed loop: alarms trigger hypothesis generation, evidence is retrieved and verified, verified diagnoses translate into maintenance actions, and each episode contributes regression tests that guard against degradation. The hierarchical verification mechanism of Stage V (Section 4) combines rule-based checking, cross-attention evidence scoring (CAEVN), and Mamdani fuzzy inference gating, with all parameters jointly optimized via MOPSO.

Recent work has shown that LLMs can encode vibration signals through fine-tuning [8], integrate domain knowledge via retrieval-augmented generation [9], and serve as reasoning copilots for prognostics and health management [10, 11]. However, deploying LLMs in industrial settings introduces new risks that are absent in traditional machine learning: hallucinated evidence that appears plausible but is factually incorrect [12], overconfident diagnoses that bypass uncertainty awareness [13], and inconsistent outputs that resist auditing [14]. These risks compound when LLM outputs directly drive maintenance actions with physical consequences in safety-critical sectors such as power grids, rotating machinery, and wind energy systems. The stakes rise further under the emerging Industry 5.0 setting, which envisions human-centric, sustainable, and resilient industrial systems [15, 16] where AI-driven diagnostics must not only be accurate but also transparent, auditable, and amenable to human oversight. Current open-loop LLM deployments cannot satisfy these requirements. Our framework addresses these I5.0 pillars directly. For *human-centricity*, a three-outcome gating mechanism routes uncertain diagnoses to human experts with interpretable fuzzy confidence scores rather than opaque classifications. For *resilience*, a regression testing loop automatically detects and rolls back performance degradations before they propagate to operations. For *sustainability*, graded gating prevents over-maintenance triggered by false-positive diagnoses, reducing unnecessary physical interventions.

Despite this progress, deploying LLMs in industrial fault diagnosis remains challenging for three interconnected reasons. First, to our knowledge, no existing framework provides an end-to-end closed loop from alarm triggering through evidence-verified diagnosis to regression-tested action. The pipeline is typically truncated at classification or, at best, at one-shot recommendation without systematic quality assurance. Second, current LLM-based diagnostic systems lack evidence-driven verification mechanisms that explicitly check hypothesis-evidence consistency and detect contradictions before committing to a diagnosis. Third, the field lacks LLMOps-grade evaluation protocols that measure not only single-episode accuracy but also longitudinal stability metrics such as regression pass rates, performance drift across updates, and rollback frequencies.

To address these challenges, we propose CL-LLMOps, a closed-loop LLMOps framework for industrial fault diagnosis that integrates five tightly coupled stages: alarm normalization, hypothesis generation, evidence-driven verification, action planning, and regression testing with automatic test set evolution. The central design choice is a hierarchical verification mechanism that combines rule-based pattern checking, a cross-attention neural verifier [17, 18], and LLM-assisted explanation. A critical constraint is that the LLM serves an explanatory rather than decisional role, ensuring that quantitative evidence assessments are not overridden by potentially hallucinated reasoning. A Mamdani

fuzzy inference system [19] replaces conventional hard-threshold gating to produce continuous action confidence scores. The gating thresholds, verifier regularization, and sampling depth are jointly tuned via multi-objective particle swarm optimization [20] that simultaneously maximizes diagnostic F1, minimizes the released-hallucination rate, and minimizes the human-handoff rate.

The main contributions of this paper are summarized as follows:

1. We propose CL-LLMOps, a closed-loop LLMOps framework with five stages, namely Alarm, Hypothesis, Verification, Action, and Regression, that delivers end-to-end fault diagnosis with version-controlled audit trails and continuous quality improvement.
2. We design a hierarchical verification mechanism integrating cross-attention evidence scoring, Mamdani fuzzy inference gating, and MOPSO at the *semantic verification level*. To our knowledge, soft computing has not previously been applied at this level. Under a paired evaluation sharing all generations between arms, the mechanism cuts the released-hallucination rate from 43.7% to 17.7% on the primary benchmark, from 24.5% to 1.5% on CWRU, and from 50.9% to 10.0% on CARE, all significant under Holm-corrected Wilcoxon tests, at a human-handoff cost that the optimizer exposes as an explicit Pareto front rather than hiding in a single number. Generic verification machinery does not reach this trade-off under the same protocol: a keyword guardrail leaves released hallucination near the unverified rate, and an off-the-shelf entailment verifier used as the gate suppresses it only by escalating every episode.
3. We demonstrate a cross-domain regression gating protocol in which parameter updates derived on one domain are replayed against the regression suites of all domains, showing on measured update cycles that the gate blocks updates whose false-escalation side effects would escape per-domain accuracy tracking.
4. We conduct experiments on three datasets, two public and one synthetic, spanning power quality, bearing fault, and wind turbine SCADA domains under a preregistered statistical protocol with stratified-bootstrap splits, paired significance tests, and multiple-comparison correction, reporting both the consistent hallucination containment and its uneven cross-domain handoff cost.

The framework employs self-consistency sampling [21] as a standard hypothesis filtering step. The contributions above target the verification, gating, and optimization layers that process the filtered hypotheses.

The remainder of this paper is organized as follows. Section 2 reviews related work in industrial fault diagnosis, LLM-based diagnostics, and LLMOps. Sections 3 and 4 formalize the problem and detail the five-stage methodology, respectively. Section 5 describes the experimental setup, Section 6 reports the results, and Section 7 concludes the paper.

2. Related Work

2.1. Industrial Fault Diagnosis and PHM

Industrial fault diagnosis has progressed through four methodological generations. Early approaches rely on hand-crafted features, including wavelet transforms, Fourier analysis, and S-transforms, fed into support vector machines or decision trees [2, 22]. While effective for well-characterized fault modes defined by standards such as IEEE 1159 [23], these methods require extensive domain expertise and generalize poorly to unseen operating conditions [7].

Deep learning eliminated the need for manual feature engineering. Wen et al. [3] proposed a one-dimensional CNN that processes raw vibration signals end-to-end with strong anti-noise properties. Recurrent models such as LSTM networks were subsequently applied to power quality disturbance classification [24, 25] and, in combination with CNN feature extractors, to bearing failure detection [26]. More recently, hybrid CNN-Transformer architectures have achieved state-of-the-art accuracy on both bearing fault [4, 5] and power quality tasks [27, 28]. For wind turbine health management, SCADA-based approaches use multivariate time-series models to detect incipient faults from operational data [29, 30, 31].

A key limitation shared by all these methods is that they produce only a fault label or a remaining useful life estimate. Among the methods surveyed above, none provides the evidence chain, confidence assessment, or actionable

maintenance guidance needed for real-world deployment. In practice, a maintenance engineer receiving a classification output still needs to verify the diagnosis against sensor evidence, assess the associated risk level, and decide on the appropriate intervention. Current methods leave these tasks entirely to human judgment.

2.2. LLMs in Industrial Operations

The application of large language models to fault diagnosis can be organized into three capability tiers. At the first tier, *fine-tuning for classification*, FD-LLM [8] demonstrated that fine-tuning Llama-3 with LoRA on FFT-encoded vibration features achieves near-perfect accuracy on the CWRU dataset. BearLLM [32] further incorporates structured prior knowledge into LLM-based bearing health management.

At the second tier, *retrieval-augmented generation for knowledge grounding*, TG-RL-RAG [9] uses reinforcement learning to train a retrieval network that eliminates manual labeling requirements. FaultGPT [33] explores vision-language models for multimodal fault diagnosis. Several recent surveys review this landscape [34, 35].

At the third tier, *agents with tool calling and planning* [36], PHM-GPT [10] proposes a domain-specific LLM covering the entire PHM lifecycle from anomaly detection to maintenance planning. UniFault [37] introduces a foundation model pretrained on 6.9 million bearing samples that generalizes across datasets without task-specific fine-tuning. Yan et al. [11] further conceptualize LLM agents as PHM copilots that coordinate diagnostic workflows.

While these methods show remarkable LLM capability for industrial tasks, Wang et al. [38] identify hallucination, value non-alignment, and weak specialization as persistent systems engineering challenges for industry-grade LLM deployment. Critically, all methods surveyed above operate in an open-loop fashion without systematic verification of outputs against empirical evidence. None of the surveyed methods incorporates regression testing that guards against quality degradation over time, and none provides a principled mechanism for deciding when to trust the LLM's diagnosis versus escalating to human review. This open-loop pattern compounds the well-documented hallucination tendencies of LLMs [12], which can yield plausible yet factually incorrect diagnoses and drive inappropriate maintenance actions in safety-critical environments. These verification and governance gaps motivate examining the broader LLM reliability literature and soft computing toolkit for techniques that could address them.

2.3. Reliable LLM Techniques

Several techniques have been developed for improving LLM reliability, yet they remain largely unapplied to industrial fault diagnosis. Self-consistency decoding [21] improves reasoning reliability by sampling multiple chain-of-thought [39] paths and selecting the most consistent answer via majority voting. Language model calibration research [13] reveals that LLMs are often poorly calibrated, with confidence scores misaligned with actual correctness probabilities. The TRUE benchmark [40] demonstrates that natural language inference classifiers can detect unsupported claims in generated text.

Soft computing approaches offer principled tools for handling the uncertainty inherent in LLM verification. Fuzzy logic [41, 19] provides interpretable decision-making under vagueness through linguistic rules and graded membership, with extensive applications in industrial control and decision support [42]. Neuro-fuzzy systems [42, 43] combine neural network learning with fuzzy linguistic interpretability, and have been applied to fault classification in rotating machinery [44, 45], power systems, and process control. However, existing neuro-fuzzy approaches target signal-level diagnosis rather than the verification of LLM-generated textual outputs. Interval Type-2 fuzzy sets [46] further extend the representational power by modeling intra-expert and inter-expert uncertainty in membership functions, which is relevant when verification scores carry measurement noise.

The growing body of work combining evolutionary optimization with fuzzy architectures for industrial fault detection [44, 47] demonstrates the potential of such hybrid approaches, though existing efforts remain confined to signal-level processing. Evolutionary optimization, including the grey wolf optimizer [48], sine cosine algorithm [49], whale optimization algorithm [50], and multi-objective particle swarm optimization [51, 20], enables joint tuning of heterogeneous system parameters under competing objectives. Sun et al. [52] recently demonstrated the effectiveness of fuzzy logic-constrained PSO for industrial design problems with non-linear constraints. Mirjalili et al. [47] review nature-inspired multi-objective techniques in engineering applications. Recent advances in adaptive parameter control for evolutionary algorithms [53] further improve convergence efficiency in high-dimensional search spaces.

A natural question is whether established soft computing classifiers, such as ANFIS for direct fault classification, fuzzy C-means clustering for anomaly segmentation, or genetic algorithm-optimized neural networks, could serve as

alternatives to the methods adopted in our framework. The key distinction is one of *abstraction level*. These methods operate on raw sensor signals to produce fault labels, whereas the verification task in CL-LLMOps operates on *textual LLM outputs* such as structured hypotheses, natural-language evidence, and semantic similarity scores, which require cross-modal reasoning between linguistic and numerical domains. An ANFIS classifier [43], for example, operates on hand-crafted features and would need to be retrained from scratch for each new fault domain, whereas a cross-attention verifier generalizes across domains by operating in the shared embedding space of a pre-trained language model. We include ANFIS as a baseline in our experiments (Section 6.1) to empirically quantify this abstraction-level gap. Similarly, while NSGA-II [54] or MOEA/D [55] could replace MOPSO for parameter optimization, MOPSO’s external archive with adaptive grid offers efficient diversity maintenance for this low-dimensional continuous problem without the overhead of non-dominated sorting [20]. Despite this rich toolkit, to our knowledge, soft computing methods have not yet been applied to the specific problem of LLM output verification in industrial machine health management settings. Our work addresses this gap by integrating cross-attention neural scoring, Mamdani fuzzy gating, and evolutionary parameter optimization into a unified verification pipeline.

On the engineering side, NeMo Guardrails [56] provides programmable safety constraints for LLM-powered applications. Toolformer [57] shows how LLMs can invoke external tools to ground their responses in factual computation. Conformal prediction [58, 59] offers distribution-free uncertainty quantification but has seen limited application to diagnostic systems. Recent surveys on LLM hallucination [12, 14] further highlight the urgency of verification in safety-critical domains.

In contrast to applying any single technique in isolation, our verification mechanism integrates self-consistency, cross-attention evidence scoring, and fuzzy inference-based gating into a unified, domain-adapted pipeline for industrial fault diagnosis. In addition, we employ multi-objective evolutionary optimization to jointly tune the fuzzy and neural parameters, bridging the gap between soft computing methodology and LLM reliability engineering.

2.4. LLMOps and Continuous Governance

The operational governance of ML systems has evolved from MLOps [60, 61] to the emerging approach of LLMOps. Sculley et al. [62] identified pervasive technical debt in ML systems, including silent failures from training-serving skew, feedback loops, and configuration drift. Continuous integration for ML [63] proposed automated testing pipelines that validate model performance before deployment. LLMOps extends these principles to handle prompt management, retrieval index maintenance, and non-deterministic outputs.

However, LLMOps has been predominantly discussed in the context of general-purpose applications such as chatbots and content generation. Its application to safety-critical industrial systems remains largely unexplored. In such systems, failures carry physical consequences and regulatory requirements demand auditability. Distinctive challenges arise in industrial LLMOps: prompt modifications can silently alter diagnostic behavior, retrieval index updates can shift evidence distributions, and non-deterministic LLM outputs resist traditional regression testing approaches that assume deterministic input-output mappings. Our regression testing methodology addresses these challenges through automatic test set evolution and version-controlled audit trails.

Table 1 summarizes the capability coverage of representative methods.

As the comparison in Table 1 reveals, no existing approach covers more than three of these seven capabilities simultaneously. In particular, evidence-driven verification, soft computing methodology for gating, and closed-loop governance with regression testing remain absent from all surveyed systems. Our work addresses these gaps by integrating all seven capabilities within a single pipeline.

3. Problem Formulation and Framework Overview

This section formalizes the industrial fault diagnosis task and introduces the CL-LLMOps framework architecture.

3.1. Problem Definition

We model each fault diagnosis task as a *diagnostic episode*. At time t , an alarm $\mathbf{a}_t$ triggers the system to analyze a multivariate observation window $\mathbf{x}_{t-w:t} \in \mathbb{R}^{w \times d}$ from sensors such as voltage, current, vibration, or temperature channels. The episode context comprises three components. The event log $\mathbf{e}$ records recent alarms, operator actions,

Table 1: Capability comparison of representative fault diagnosis methods. ✓ indicates the capability is present; ✗ indicates it is absent. Soft Comp. is marked ✓ if the method uses fuzzy logic, neuro-fuzzy inference, or evolutionary parameter optimization in any pipeline component, regardless of whether it is signal-level or semantic-level. Closed-Loop requires regression testing or automatic rollback on top of versioned deployment.

Method	Year	Signal Proc.	DL Classif.	LLM Diag.	Evidence Verif.	Action Plan.	Soft Comp.	Closed- Loop
SVM/DT classical	2018–20	✓	✗	✗	✗	✗	✗	✗
ANFIS [43]	1993	✓	✗	✗	✗	✗	✓	✗
1D-CNN [3]	2018	✓	✓	✗	✗	✗	✗	✗
CNN-Transformer [27]	2025	✓	✓	✗	✗	✗	✗	✗
FD-LLM [8]	2025	✓	✗	✓	✗	✗	✗	✗
BearLLM [32]	2025	✓	✗	✓	✗	✗	✗	✗
PHM-GPT [10]	2025	✓	✗	✓	✗	✓	✗	✗
TG-RL-RAG [9]	2025	✓	✗	✓	✗	✗	✗	✗
NeMo Guardrails [56]	2023	✗	✗	✗	✓	✗	✗	✗
MLOps frameworks [60]	2023	✗	✓	✗	✗	✗	✗	✓
CL-LLMOps (Ours)	2026	✓	✓	✓	✓	✓	✓	✓

and state transitions. Asset metadata $\mathbf{m}$ describes equipment type, operational mode, and maintenance history. An optional domain knowledge base $\mathcal{K}$ supplies fault catalogs and standard operating procedures.

The system produces a structured output $(y, \mathcal{H}, \mathcal{E}, \mathbf{u}, \mathcal{T})$ that extends conventional classification along three axes. It provides ranked hypotheses $\mathcal{H} = \{h_1, \dots, h_K\}$ with evidence chains $\mathcal{E}(h_k)$ for each, actionable maintenance recommendations $\mathbf{u}$ with risk assessment, and regression test artifacts $\mathcal{T}$ for continuous quality assurance. The action space is discretized into five escalation levels:

$$\mathcal{U} = \{\text{A0: monitor, A1: recalibrate, A2: repair, A3: shutdown, A4: emergency}\}. \quad (1)$$

This five-level scheme abstracts over domain-specific interventions to enable cross-domain evaluation. In deployment, each level expands to a domain-specific action catalog where the LLM generates remediation procedures covering affected components, required tools, and shutdown constraints.

3.2. Framework Overview

We organize the diagnostic process into five stages that form a closed loop (Fig. 1).

Stage A (Alarm Normalization). Industrial sensors produce heterogeneous signals such as three-phase voltage waveforms, high-frequency vibration data, and low-rate SCADA telemetry. These signals differ in sampling rate, scale, and semantics across domains. Stage A transforms these raw readings and alarm signals into a structured evidence pack through anomaly detection, statistical summarization, and feature extraction. Time-domain statistics such as RMS, kurtosis, and crest factor are combined with frequency-domain features including FFT peaks and envelope spectra. The resulting evidence pack provides a domain-agnostic interface that allows all downstream stages to operate without domain-specific prompt engineering, as detailed in Section 4.1.

Stage H (Hypothesis Generation). Given the evidence pack, the LLM agent generates candidate root-cause hypotheses $\mathcal{H}$ by repeated sampling. Each hypothesis is a structured record specifying the diagnosed fault mode, its root-cause mechanism, a severity rating, the evidence cited from the alarm context, and recommended inspection points. To suppress hallucinated hypotheses that appear only sporadically, Stage H applies self-consistency sampling: the LLM is queried M times and the hypothesis most consistent with the sample population, measured in sentence-embedding space, is selected for verification. This selection step is critical because LLM agents can generate plausible-sounding but factually incorrect fault explanations, particularly for rare or ambiguous alarm patterns. The full hypothesis generation procedure is described in Section 4.2.

Stage V (Verification). Verification is the core technical contribution of this work. A hierarchical mechanism evaluates each hypothesis against retrieved evidence through three layers: rule-based pattern matching that enforces

Table 2: Structured alarm schema across three industrial domains. Each alarm is normalized into a unified format for consistent LLM input.

Field	Power Quality	Bearing (CWRU)	Wind Turbine
alarm_type	PQ disturbance	Vibration anomaly	SCADA exceedance
timestamp	Cycle ($T=20\text{ms}$)	Index at 12 kHz	10-min interval
severity	IEEE 1159 class	Amplitude deviation	Priority code
channels	V_{abc}, I_{abc}	DE & FE accel.	52 SCADA tags
window_stats	RMS, THD, crest	Mean, std, kurtosis	Mean, std, trend
freq_feat.	FFT, harmonic order	Envelope peaks	N/A (low-rate)
asset_info	Bus ID, rating	Motor HP, speed	Model, rated power
context	Topology, load	Load (0–3 HP)	Wind speed, temp.

necessary physical conditions, a cross-attention neural verifier that scores hypothesis–evidence consistency, and LLM-assisted explanation that generates audit trails. A Mamdani fuzzy inference gating strategy maps the verification results to one of three outcomes: proceed, escalate, or defer. This produces smooth decision transitions rather than the abrupt flips of hard thresholds. This design constrains the LLM to an interpretive support function, preventing hallucinated reasoning from overriding quantitative evidence. The complete verification mechanism is formalized in Section 4.3.

Stage A₂ (Action Planning). Accurate diagnosis alone is insufficient for industrial settings; the system must also recommend appropriate maintenance actions. Stage A₂ translates verified diagnoses into actions from $\mathcal{U}$ (Eq. (1)), considering fault severity, risk level, and operational context to generate step-by-step remediation procedures. A safety constraint couples verification confidence to action authority: high-risk interventions such as shutdown or emergency actions require stronger evidence confidence, preventing the system from recommending aggressive measures based on weakly supported diagnoses. Details appear in Section 4.4.

Stage R (Regression Testing). Industrial LLM systems face a silent degradation problem: modifications to prompts, retrieval indices, or model weights can inadvertently break previously correct diagnoses without any visible error signal. Stage R addresses this by generating test cases from each completed episode that capture input conditions, expected outputs, and evaluation criteria. The test set grows automatically through four channels: escalated cases, verifier–LLM disagreements, high-divergence episodes, and synthetic perturbations. All configurable components are version-controlled, including prompts, retrieval indices, verifier weights, and action templates. New versions must pass the full regression suite before deployment. This continuous quality assurance mechanism is described in Section 4.6.

The closed-loop design distinguishes CL-LLMOps from one-shot pipelines: Stage R continuously expands the test suite and triggers retraining or prompt refinement when regression failures occur, ensuring reliability across updates and distributional shifts.

Table 2 shows how alarm normalization maps domain-specific data into a unified schema. Despite the heterogeneity across three-phase electrical waveforms, high-frequency vibration signals, and low-rate SCADA telemetry, the structured alarm pack provides a consistent interface that allows the downstream LLM agent to operate without domain-specific prompt engineering. We include both time-domain statistics and frequency-domain features as pre-computed evidence, reducing the LLM’s burden of performing numerical computation on raw signals.

4. Methodology: Closed-loop LLMOps for Fault Diagnosis

This section details the five stages of the CL-LLMOps framework, with Stage V receiving the most extensive treatment as it houses the main technical contribution.

4.1. Stage A: Alarm Normalization

We convert heterogeneous raw sensor data and alarm signals into a structured evidence pack $\mathcal{P}_t$. Each alarm window at time t enters a feature extraction step that computes time-domain statistics and frequency-domain features. The features are then combined with contextual information into a single document:

$$\mathcal{P}_t = \{\text{alarm} : \mathbf{a}_t, \text{features} : \mathbf{f}_t, \text{context} : (\mathbf{e}, \mathbf{m}), \text{version} : v\} \quad (2)$$

where $\mathbf{f}_t$ denotes the extracted feature vector and v is a schema version identifier that enables backward compatibility across system updates.

4.2. Stage H: Hypothesis Generation

Given the evidence pack $\mathcal{P}_t$, the LLM agent generates candidate root-cause hypotheses by self-consistency sampling following Wang et al. [21]: the same prompt is decoded M times independently at temperature $T=0.3$, where M is tuned jointly with the gating parameters ($M=7$ at the deployed operating point, Section 4.5). The prompt template, reproduced verbatim in Appendix A of the supplementary material, enforces a structured JSON output whose keys are fixed by the downstream parser: the diagnosed fault mode, the root-cause mechanism, a five-level severity rating aligned with the action rule table, the evidence cited from the alarm features and knowledge base, recommended inspection points, and cascading effects if unaddressed. We set $T=0.3$ following the self-consistency literature, which recommends moderate temperatures to balance hypothesis diversity against output quality. Lower temperatures preserve the structured JSON format required for downstream processing while still producing diverse reasoning paths. Each sample h_m is embedded as $\mathbf{e}_m$ by the sentence encoder, and the selected hypothesis is the medoid of the sample population,

$$h^* = \arg \max_{h_m} \cos(\mathbf{e}_m, \bar{\mathbf{e}}), \quad c = \binom{M}{2}^{-1} \sum_{m < m'} \cos(\mathbf{e}_m, \mathbf{e}_{m'}), \quad (3)$$

where $\bar{\mathbf{e}}$ is the embedding centroid. The consistency score c summarizes inter-sample agreement; it is recorded with the episode and exposed to the substitute verification baselines of the ablation study, while the fuzzy gate itself consumes only the Stage V signals below. Sporadic hallucinated hypotheses are distant from the centroid and are therefore never selected.

4.3. Stage V: Evidence-Driven Verification

The key challenge in deploying LLM agents for industrial fault diagnosis is *hallucination*. We address this by treating each hypothesis as an unverified claim that must be tested against empirical evidence (*verify before acting*). V1 retrieves targeted evidence, V2 scores hypothesis–evidence consistency, and V3 gates the diagnosis into proceed, escalate, or defer.

4.3.1. V1: Evidence Retrieval

The selected hypothesis h^* , abbreviated h below, queries the domain knowledge base $\mathcal{K}$ by semantic search: the top- J ($J=5$) most similar entries to the hypothesis embedding are retrieved from a FAISS inner-product index. $\mathcal{K}$ holds three channels per dataset: curated textbook fault signatures encoding the physical conditions of Table 3, historical case reports drawn from the training split, and feature-statistics summaries; every entry carries its fault-class tag and provenance. We denote the retrieved evidence set as $\mathcal{E}(h) = \{e_1, \dots, e_J\}$.

4.3.2. V2: Hierarchical Consistency Verification

Layer 1: Rule-based verification. Deterministic class-consistency predicates $r : (h, e_j) \rightarrow \{0, 1\}$ check whether each retrieved evidence item’s fault-class tag agrees with the diagnosed fault mode parsed from h ; the physical conditions that ground these tags (Table 3) are encoded in the curated signature texts of $\mathcal{K}$, so a hypothesis whose diagnosis contradicts the retrieved signatures receives no rule support. The layer outputs the support fraction $v_{\text{rule}}(h) = \frac{1}{J} \sum_j r(h, e_j) \in [0, 1]$; a hypothesis with $v_{\text{rule}}=0$ is fully unsupported by Layer 1.

Layer 2: Cross-attention evidence verifier. We employ a *cross-attention evidence verification network* (CAEVN) [18, 17] that scores how well the retrieved evidence supports the hypothesis. Let $\mathbf{h} \in \mathbb{R}^d$ denote the hypothesis embedding and $\mathbf{e}_j \in \mathbb{R}^d$ the j -th evidence embedding. Multi-head cross-attention treats $\mathbf{h}$ as the query and $\{\mathbf{e}_1, \dots, \mathbf{e}_J\}$ as keys and values:

$$\mathbf{c}(h) = \sum_{j=1}^J \alpha_j \mathbf{V} \mathbf{e}_j, \quad (4)$$

$$\alpha_j = \frac{\exp(\mathbf{Q} \mathbf{h} \cdot \mathbf{K} \mathbf{e}_j / \sqrt{d_k})}{\sum_{l=1}^J \exp(\mathbf{Q} \mathbf{h} \cdot \mathbf{K} \mathbf{e}_l / \sqrt{d_k})}$$

Table 3: Representative physical conditions encoded in the knowledge-base signature texts that ground the Layer-1 class-consistency check.

Domain	Rule	Condition
PQ	Sag detection	RMS < 0.9 p.u. for ≥ 1 cycle
PQ	Harmonic ID	THD > 5% & dominant order match
PQ	Swell detection	RMS > 1.1 p.u. for ≥ 1 cycle
CWRU	Inner race	BPFI peak within $\pm 3\%$ tolerance
CWRU	Outer race	BPFO peak within $\pm 3\%$ tolerance
CWRU	Ball fault	BSF peak within $\pm 5\%$ tolerance
CARE	Temp. anomaly	$\Delta T_{\text{gen}} > 2\sigma$ from baseline
CARE	Power drop	$P_{\text{actual}}/P_{\text{rated}} < 0.7$
CARE	Multi-signal	≥ 2 correlated deviations

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{d_k \times d}$ are learned projections and the full model uses $H=4$ parallel heads ($d_k=96$). The attended context passes through a two-layer GELU feed-forward head to a sigmoid support score:

$$g_\phi(h, \mathcal{E}) = \sigma(\mathbf{w}_o^\top \text{FFN}(\mathbf{c}(h)) + b_o) \in [0, 1]. \quad (5)$$

Scoring the full set yields the aggregate support $g_\phi(h, \mathcal{E}(h))$; scoring each item alone, $g_\phi(h, \{e_j\})$, yields per-item verdicts used for contradiction counting and diagnosis tracing. Training uses AdamW with cosine annealing for 50 epochs on a per-dataset triple set of support/contradict pairs generated from the training split.

Layer 3: LLM-assisted explanation. After the gate has fixed its decision, the LLM produces a human-readable explanation of the quantitative verdict—conditioned on the retrieved evidence, the layer scores, and the decision itself—which is recorded in the episode audit log but *not* fed back into the gating score (Eq. (9)), ensuring hallucinated reasoning cannot override quantitative evidence assessment.

We combine rule-based and neural verification through a weighted average. Rules catch mandatory class-consistency constraints while the neural verifier captures softer patterns; the weight λ_r balances strictness against flexibility. The verification score for hypothesis h is

$$s(h) = \text{clip}_{[0,1]}[\lambda_r \cdot v_{\text{rule}}(h) + (1 - \lambda_r) \cdot g_\phi(h, \mathcal{E}(h))] \quad (6)$$

where $v_{\text{rule}} \in [0, 1]$ is the Layer-1 support fraction, g_ϕ is the set-level support score of Eq. (5), and $\lambda_r \in [0, 1]$ controls the trade-off. Two further gate inputs come from the same layers: the contradiction severity $c(h) = \frac{1}{J} |\{e_j : g_\phi(h, \{e_j\}) < 0.5\}|$, the fraction of evidence items whose per-item verdict opposes the hypothesis, and the evidence coverage $\rho(h)$, the mean normalized cosine relevance of the retrieved evidence to the hypothesis embedding, which is low precisely when the knowledge base holds nothing close to the claimed fault mode.

4.3.3. V3: Fuzzy Inference-Based Gating

We employ a Mamdani-type *fuzzy inference system* (FIS) [19, 41] that models the gating decision as a smooth, interpretable mapping from *meta-cognitive inputs* to an action confidence score. The gating layer must satisfy two requirements that jointly motivate a rule-based fuzzy formulation over a calibrated scalar mapping such as Platt or isotonic post-processing. First, plant engineers must be able to audit and amend the decision logic without retraining: the eleven linguistic rules in Table 4 expose the gating policy as human-readable IF–THEN statements over verification score, evidence coverage, and contradiction severity, whereas a fitted parametric curve offers no handle for encoding site-specific safety policy such as “dense but contradictory evidence must escalate.” Second, the gate implements a genuinely three-way decision semantics in which *escalate* is a first-class outcome occupying the graded middle region between proceed and defer, rather than a band carved post hoc out of a one-dimensional probability. The empirical comparison against calibrated probabilistic gating appears in Section 6.2; our claim is not that fuzzy inference dominates well-calibrated scalar gating on aggregate safety metrics, but that it meets the auditability and three-way-semantics requirements above, with the aggregate safety cost quantified in Section 6.2. We select Mamdani inference for its interpretability [64, 65, 42]. We use Type-1 fuzzy sets [66, 46]; the three-input dimensionality makes Type-2’s added complexity unwarranted.

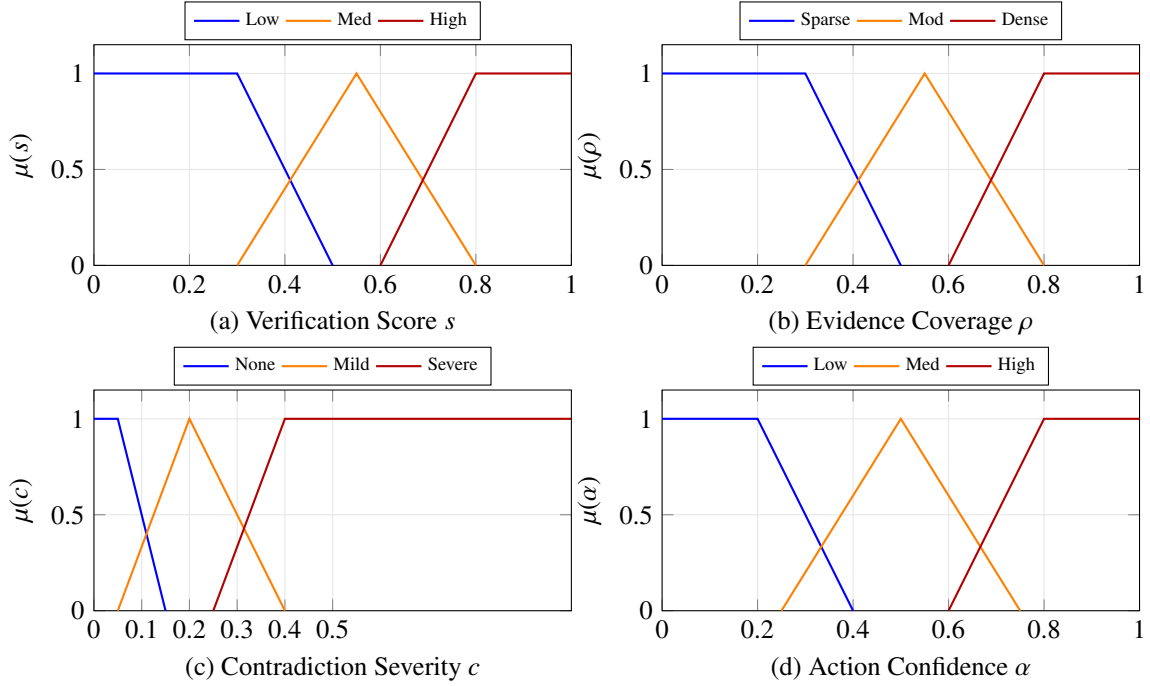


Figure 2: Membership functions for the Mamdani FIS. (a)–(c) Input variables; (d) output variable. Breakpoints are expert-set and robustness-tested under simultaneous perturbation (Section 6.3).

Input linguistic variables.. Three input variables are each partitioned into three terms using trapezoidal (trap) or triangular (tri) membership functions. (i) *Verification Score* $s \in [0, 1]$:

$$\mu_L^s = \text{trap}(s; 0, 0, 0.3, 0.5), \quad \mu_M^s = \text{tri}(s; 0.3, 0.55, 0.8), \\ \mu_H^s = \text{trap}(s; 0.6, 0.8, 1, 1).$$

(ii) *Evidence Coverage* $\rho \in [0, 1]$, with terms **Sparse**, **Mod**, **Dense** and breakpoints identical to the verification score.

(iii) *Contradiction Severity* $c(h) \in [0, 1]$ as defined in Section 4.3.2:

$$\mu_{\text{None}}^c = \text{trap}(c; 0, 0, 0.05, 0.15), \quad \mu_{\text{Mild}}^c = \text{tri}(c; 0.05, 0.2, 0.4), \\ \mu_{\text{Severe}}^c = \text{trap}(c; 0.25, 0.4, 1, 1).$$

Breakpoints are expert-set; the gating behavior remains stable under simultaneous ± 0.10 breakpoint perturbation (Section 6.3), while the gate thresholds and weights are tuned by MOPSO (Section 4.5). The partition is deliberately non-Ruspini, creating an ambiguity zone that maps to escalation behavior (Fig. 2).

Fuzzy rule base.. Table 4 presents eleven IF–THEN rules. The output *Action Confidence* $\alpha \in [0, 1]$ uses terms {Low, Med, High} with $\mu_L^\alpha = \text{trap}(\alpha; 0, 0, 0.2, 0.4)$, $\mu_M^\alpha = \text{tri}(\alpha; 0.25, 0.5, 0.75)$, $\mu_H^\alpha = \text{trap}(\alpha; 0.6, 0.8, 1, 1)$.

Rule derivation methodology.. The rules were derived in three steps. First, domain standards including IEEE 1159 [23], ISO 10816 [67], and IEC 61400 [68] define safety requirements that map to R10–R11. Second, the remaining rules encode standard maintenance-practice heuristics relating evidence support, coverage, and contradiction to release confidence. Third, wildcard notation compresses the $3^3 = 27$ input combinations into 11 rules. Exhaustive enumeration confirms that the 11 rules cover all $3^3 = 27$ antecedent combinations: R10 subsumes 9 cells where Score=Low, R11 adds 6 cells with Severe contradictions excluding Low-score cases, and R1–R9 cover the remaining 12 cells. When multiple rules match the same cell, the max-aggregation operator in Eq. (7) resolves conflicts by selecting the dominant consequent. All 11 rules fire simultaneously via Eq. (7).

Table 4: Fuzzy rule base for the gating mechanism. “*” denotes any linguistic value.

Rule	Score	Coverage	Contra.	→ Confidence
R1	High	Dense	None	High
R2	High	Dense	Mild	Medium
R3	High	Moderate	None	High
R4	High	Sparse	*	Low
R5	Medium	Dense	None	Medium
R6	Medium	Moderate	None	Medium
R7	Medium	*	Mild	Low
R8	Medium	Sparse	None	Low
R9	High	Moderate	Mild	Medium
R10	Low	*	*	Low
R11	*	*	Severe	Low

Inference and defuzzification. For each rule R_i with antecedent terms $A_i^{(1)}$, $A_i^{(2)}$, $A_i^{(3)}$ and consequent B_i , the firing strength is:

$$w_i = \min(\mu_{A_i^{(1)}}(s), \mu_{A_i^{(2)}}(\rho), \mu_{A_i^{(3)}}(c)) \quad (7)$$

The T-Min t-norm ensures no compensation between antecedent terms, a conservative property desirable in safety-critical gating. The aggregated output membership function $\mu_{\text{agg}}(\alpha) = \max_i \min(w_i, \mu_{B_i}(\alpha))$ yields the action confidence via centroid defuzzification:

$$\alpha(h) = \frac{\int_0^1 \alpha \cdot \mu_{\text{agg}}(\alpha) d\alpha}{\int_0^1 \mu_{\text{agg}}(\alpha) d\alpha} \quad (8)$$

The final gating decision maps the defuzzified confidence to one of three outcomes, with two safety overrides evaluated first:

$$\text{Decision}(h) = \begin{cases} \text{defer} & \text{if } \rho(h) < \theta_p \\ \text{escalate} & \text{if } \mu_{\text{Severe}}^c(c(h)) \geq 0.5 \\ \text{proceed} & \text{if } \alpha(h) \geq \alpha_p \\ \text{defer} & \text{if } \alpha(h) < \alpha_d \\ \text{escalate} & \text{otherwise} \end{cases} \quad (9)$$

where α_p and α_d are the proceed and defer thresholds and θ_p is the coverage-defer threshold. The first case defers insufficient-evidence episodes; the second implements the site-safety policy that densely contradicted hypotheses must reach a human regardless of aggregate confidence. For high-risk hypotheses, whose parsed severity is high or critical and whose rule-mapped actions fall in $\{A3, A4\}$, we raise α_p by a safety margin δ . The three-outcome design separates escalate from defer, communicating distinct failure modes to operators. Escalated and deferred episodes are routed to human review and added to the regression test set. Algorithm 1 summarizes the complete procedure, and Fig. 3 illustrates the three-layer verifier with the Mamdani FIS gating module that routes each episode to *proceed*, *escalate*, or *defer*.

4.4. Stage A2: Action Planning

Once a hypothesis h^* passes verification, the LLM translates the diagnosis into a structured action plan containing the action level $u \in \mathcal{U}$ from Eq. (1), remediation procedures, and conditional triggers for escalation. The action level is set by mapping fault type and severity to a candidate level. A safety constraint then couples verification confidence to action authority:

$$u_{\text{allowed}} = \begin{cases} u_{\text{cand}} & \text{if } \alpha(h^*) \geq \alpha_p + \delta \cdot \mathbb{I}[\text{risk}(u) \in \{\text{high}, \text{critical}\}] \\ \min(u_{\text{cand}}, A2) & \text{otherwise} \end{cases} \quad (10)$$

This coupling prevents aggressive interventions based on weakly supported diagnoses, downgrading to at most A2 when evidence is insufficient.

Algorithm 1 Evidence-Driven Verification with Fuzzy Gating

Require: Selected hypothesis h (Eq. (3)), knowledge base $\mathcal{K}$, verifier g_ϕ , FIS $\mathcal{R}_F$, thresholds $\alpha_p, \alpha_d, \delta, \theta_p$

Ensure: Gate decision for h with audit record

```
1:  $\mathcal{E}(h) \leftarrow \text{RETRIEVETOPJ}(h, \mathcal{K})$  {FAISS semantic search}
2:  $\rho(h) \leftarrow \frac{1}{J} \sum_j \text{relevance}(\mathbf{h}, \mathbf{e}_j)$  {evidence coverage}
3:  $v_{\text{rule}}(h) \leftarrow \frac{1}{J} \sum_j r(h, e_j)$  {Layer 1: class consistency}
4:  $g \leftarrow g_\phi(h, \mathcal{E}(h))$  {Layer 2: set-level support}
5:  $c(h) \leftarrow \frac{1}{J} \|\{e_j : g_\phi(h, \{e_j\}) < 0.5\}\|$  {per-item contradictions}
6:  $s(h) \leftarrow \text{clip}_{[0,1]}[\lambda_r v_{\text{rule}}(h) + (1 - \lambda_r) g]$  {Eq. (6)}
7:  $\alpha(h) \leftarrow \text{FUZZYINFER}(s(h), \rho(h), c(h), \mathcal{R}_F)$  {Eqs. (7)–(8)}
8:  $\alpha_{\text{th}} \leftarrow \alpha_p + \delta \cdot \mathbb{I}[\text{severity}(h) \in \{\text{high}, \text{critical}\}]$  {high-risk shift}
9: if  $\rho(h) < \theta_p$  then
10:   decision  $\leftarrow$  defer {insufficient evidence}
11: else if  $\mu_{\text{Severe}}^\epsilon(c(h)) \geq 0.5$  then
12:   decision  $\leftarrow$  escalate {contradiction override}
13: else if  $\alpha(h) \geq \alpha_{\text{th}}$  then
14:   decision  $\leftarrow$  proceed
15: else if  $\alpha(h) < \alpha_d$  then
16:   decision  $\leftarrow$  defer
17: else
18:   decision  $\leftarrow$  escalate {routed to human review}
19: end if
20:  $\text{LLMEXPLAIN}(h, \mathcal{E}(h), s(h), \text{decision})$  {Layer 3: audit log, post-decision}
21: return decision,  $\mathcal{E}(h)$ ,  $s(h)$ ,  $\alpha(h)$ 
```

4.5. Joint Parameter Optimization via Multi-Objective PSO

The fuzzy MF parameters, verifier hyperparameters, and gating thresholds interact and must be jointly optimized. We formulate this as a multi-objective problem solved by *multi-objective particle swarm optimization* (MOPSO) [20, 51], selected over NSGA-II [54], MOEA/D [55], and multi-objective GWO [48] for its efficient Pareto front maintenance in low-dimensional continuous spaces.

The decision vector comprises six parameters: the gate thresholds α_p and α_d , the high-risk margin δ , the rule–neural weight λ_r , the self-consistency count M , and the CAEVN entropy-regularization weight β , the latter resolved to a grid of pre-trained checkpoints. Low-sensitivity quantities, namely the membership breakpoints, the coverage threshold θ_p , and the CAEVN architecture, are expert-fixed following adaptive parameter selection principles [53]. MOPSO (swarm size 50) simultaneously minimizes:

$$\min_{\mathbf{x}} (-F_1(\mathbf{x}), \text{Halluc}(\mathbf{x}), \text{Handoff}(\mathbf{x})) \quad (11)$$

where $\text{Handoff} = 1 - \text{proceed rate}$ is the fraction of episodes routed to a human. The third objective must price autonomy itself: penalizing only *false* escalations leaves “escalate everything” Pareto-optimal, since a gate that never proceeds trivially achieves zero hallucination while classification F_1 falls back to the alarm prediction. Pricing every handoff makes the safety–autonomy trade-off non-degenerate; the false-escalation rate remains a reported diagnostic. Feasibility requires $\alpha_p > \alpha_d + \delta$ and $\alpha_p + \delta \leq 0.95$, so high-risk episodes remain gateable. From the Pareto-optimal archive we select the solution closest to the normalized ideal point [54] (Section 6.3).

Computational complexity. The per-episode complexity is $O(M \cdot T_{\text{LLM}} + K \cdot J \cdot T_{\text{tool}} + K \cdot J \cdot T_{\text{CAEVN}})$ and is dominated by the M LLM decodes; measured end-to-end latencies appear in Section 6.4. The MOPSO optimization adds a one-time offline cost of 5,050 pipeline evaluations per run on cached validation generations. The grid-search alternative evaluated in Section 6.3 pins the operating point to the best corner of a coarse grid over $(\alpha_p, \alpha_d, \delta, \lambda_r)$, since exhaustive enumeration of the full six-dimensional space at comparable resolution is computationally prohibitive.

4.6. Stage R: Regression Testing and Continuous Improvement

Modifications to prompts, retrieval indices, or model weights can silently break previously correct diagnoses. This stage operationalizes the *resilience* pillar of Industry 5.0 [69] by maintaining an ever-growing suite of validated test cases across three dimensions.

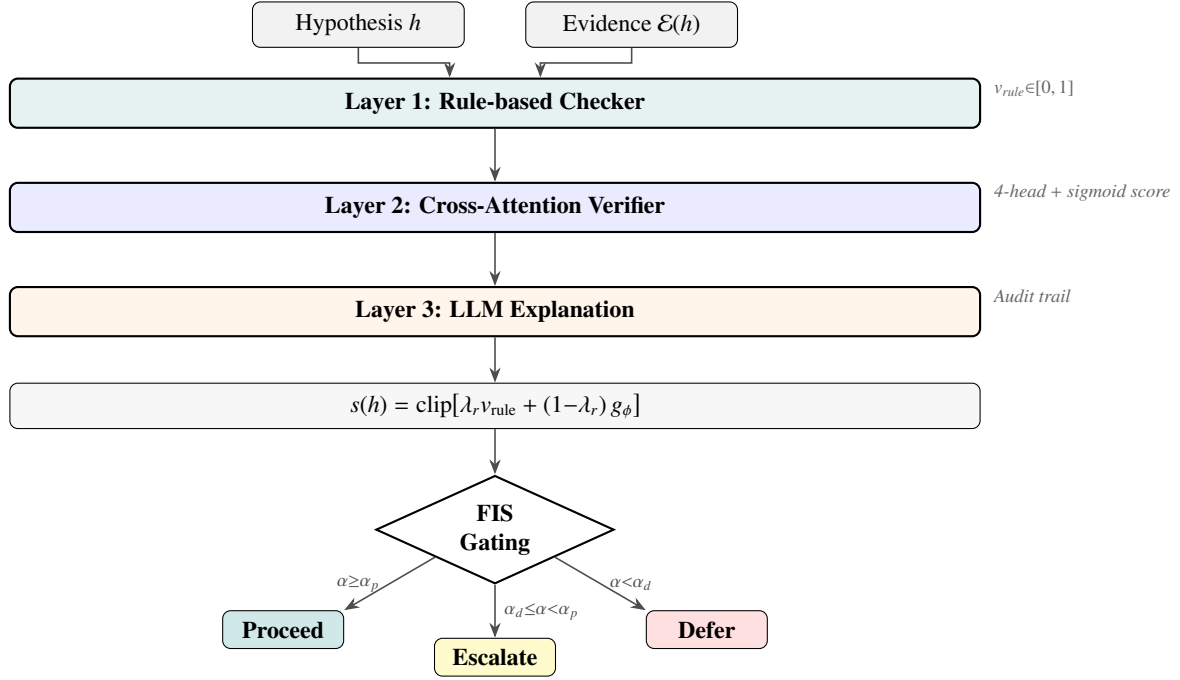


Figure 3: Hierarchical evidence-driven verifier. Layer 1 applies domain rules as necessary conditions. Layer 2 scores evidence–hypothesis alignment via cross-attention NLI. Layer 3 generates explanations for audit. The Mamdani FIS gating module routes each episode to proceed, escalate, or defer based on the aggregated confidence score α .

Each completed diagnostic episode produces a test case

$$\tau = (\mathcal{P}_t, \mathcal{Y}_{\text{allowed}}, \mathcal{E}_{\text{required}}, \mathcal{U}_{\text{allowed}}, \mathcal{U}_{\text{forbidden}}, \text{eval_script})$$

containing the frozen evidence pack, acceptable labels, required evidence types, permissible and prohibited actions, and a constraint checker. The test set $\mathcal{T}$ grows through escalated episodes, verifier–LLM disagreements, high-divergence cases, and synthetic perturbations. A new version v_{n+1} is promoted only if:

$$\text{pass_rate}(v_{n+1}) = \frac{|\{\tau \in \mathcal{T}_n : \text{Eval}(\tau, v_{n+1}) = \text{pass}\}|}{|\mathcal{T}_n|} \geq \theta_{\text{reg}} = 0.95 \quad (12)$$

Fig. 4 depicts the version governance flow: every system update is gated by the regression suite, with failed updates triggering automatic rollback and the test suite continuously expanding from escalated episodes, verifier–LLM disagreements, perturbations, and boundary cases. The five stages above form the complete CL-LLMOps framework.

5. Experimental Setup

5.1. Datasets

We evaluate CL-LLMOps on three datasets spanning distinct industrial domains to assess the generality of the closed-loop approach: one synthetic benchmark (PQ) and two public datasets (CWRU and CARE). Table 5 summarizes the key characteristics of each dataset.

Power Quality (PQ) Classification Dataset. This is the primary evaluation benchmark, reflecting the paper’s focus on power grid fault diagnosis. The dataset contains 240,000 labeled samples covering 16 categories of power quality disturbances defined by the IEEE 1159 standard, including voltage sags, swells, harmonics, transients, flicker, and their combinations. Each sample is a single- or multi-cycle waveform segment of three-phase voltage and current signals, generated by IEC-compliant parametric models with additive Gaussian noise. The benchmark provides a

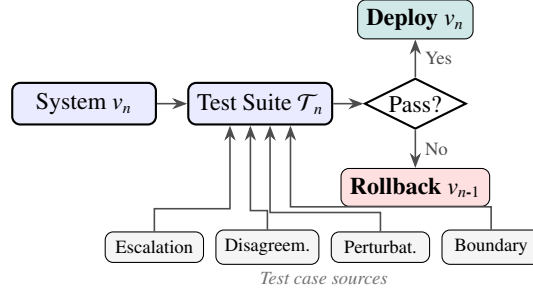


Figure 4: LLMOps version governance pipeline. Each system update must pass the full regression test suite before deployment. Failed updates trigger automatic rollback. The test suite grows through escalated episodes, verifier–LLM disagreements, synthetic perturbations, and boundary cases.

controlled testbed for cross-method comparison under the IEEE 1159 taxonomy; phenomena such as transformer saturation and ferroresonance are natural extensions for future field-data studies. We adopt a 70/15/15 train/validation/test split stratified by disturbance category. The alarm normalization stage (Stage A) computes RMS, total harmonic distortion (THD), crest factor, and FFT spectral peaks as the structured feature set.

CWRU Bearing Fault Dataset. The Case Western Reserve University bearing dataset is the de facto standard benchmark for rotating machinery fault diagnosis. It provides vibration acceleration signals sampled at 12 kHz from drive-end and fan-end accelerometers under four motor load conditions ranging from 0 to 3 HP. Three single-point fault types of inner race, outer race, and rolling element are artificially introduced via electrical discharge machining at three severity levels of 0.007, 0.014, and 0.021 inches, yielding 10 classes: one normal condition and nine fault conditions. We segment the continuous signals into fixed-length windows of 2,048 samples with 50% overlap and then apply a 70/15/15 split over windows, so overlapping windows from the same recording can fall in different partitions. In addition, the release file for the 2 HP normal baseline also contains the 1 HP recording, and our loader read the 1 HP array from it. The normal class therefore contains three distinct recordings, and 97 of the 883 test windows, all normal, are identical to training windows. The alarm normalization stage extracts time-domain statistics such as mean, standard deviation, kurtosis, RMS, and crest factor, together with envelope spectrum features at the four characteristic bearing defect frequencies.

CARE Wind Turbine SCADA Dataset. The CARE to Compare dataset comprises 95 SCADA time-series datasets from 36 wind turbines across three wind farms, spanning 89 turbine-years of operation. In the v6 release used here, 45 datasets contain labeled anomaly periods that precede actual turbine faults, while 50 represent normal behavior. Each data point includes vendor-averaged SCADA channels such as active power, rotor speed, generator temperature, pitch angle, and nacelle wind speed, recorded at 10-minute intervals; we retain 52 channels selected by non-zero availability across the three farms. We formulate the task as binary anomaly detection between normal and pre-fault states, using a sliding window of 144 time steps spanning 24 hours, with a 70/15/15 split at the dataset level: each of the 95 datasets falls in one partition. A turbine can contribute several datasets, and 6 of the 10 test turbines also contribute training datasets. The alarm normalization stage computes rolling mean, standard deviation, range, and trend slope over the retained channels. After windowing, this train/validation/test split yields 267,569 24-hour episodes in total (192,165/40,658/34,746), as summarized in Table 5.

All three datasets are controlled benchmarks rather than in-service industrial recordings: the PQ waveforms are generated from IEC-compliant parametric models, the CWRU faults are induced by electrical discharge machining, and the CARE task is formulated as binary pre-fault detection. We select them because they are reproducible and directly comparable with prior work. Because the CWRU and CARE test sets share recordings or turbines with the training data, the classification accuracies on these two datasets are optimistic estimates for unseen equipment. The results in Section 6 should therefore be read as benchmark evidence for the verification and governance mechanisms; validation on in-service equipment with naturally occurring faults and full maintenance records remains future work, as discussed in Section 7.

Table 5: Summary of evaluation datasets. The PQ Classification dataset serves as the primary benchmark (power grid domain), while CWRU and CARE validate cross-domain generalization.

Attribute	PQ Class.	CWRU	CARE Wind
Domain	Power grid	Bearing	Wind turbine
Signal type	Voltage/current	Vibration	SCADA tags
Sampling rate	3.2 kHz	12 kHz	10-min
# Classes	16	10	2 (anomaly)
# Samples	240,000	5,886	267,569
Input dim.	6 channels	2 channels	52 tags
Feature set	RMS, THD, FFT	Stats, env.	Rolling stats
Fault types	Sag, swell, harmonics, etc.	IR, OR, RE (3 sizes)	Pre-fault anomaly

5.2. Comparison Arms

The evaluation is organized around a strictly paired design that isolates the contribution of each proposed component rather than a horse race between unrelated systems.

Primary paired baseline: Agent-noV. An LLM agent identical to CL-LLMOps in perception, evidence retrieval, hypothesis generation, and action planning, but without the hierarchical verification mechanism and gating strategy: every generated hypothesis is released and acted upon. In the statistical protocol both arms share each alarm, each sampled hypothesis set, and each retrieved evidence item, so the comparison is exactly paired and any difference is attributable to verification and gating.

Verification-layer baselines. To test whether the fuzzy gate outperforms simpler gating back-ends operating on the same verification signals, we evaluate: calibrated probability gating, which replaces the Mamdani FIS with Platt or isotonic calibration of the verification score fitted on the validation split only, with the proceed/escalate coverage matched to the native gate on validation; a static safety-guardrail filter representative of keyword- and pattern-level LLM guardrails; and an off-the-shelf NLI entailment verifier used directly as the gate. These arms share the perception layer and LLM generations with CL-LLMOps.

Optimizer baseline. NSGA-II [54] replaces MOPSO for the joint parameter optimization under an identical evaluation budget, and a coarse grid-search corner point replaces the MOPSO-selected operating point in the ablation study.

Backbone baseline. Mistral-7B-Instruct-v0.3 [70] replaces Llama-3.1-8B-Instruct with all other components frozen, testing the backbone-agnostic claim.

Classical and deep classification baselines. SVM with an RBF kernel, XGBoost, ANFIS [43], a 1D-CNN, and a Transformer encoder are trained on the repository splits and evaluated on the full held-out test partitions (36,001/883/34,746 samples) in a single deterministic run (seed 42); the SVM training set is capped as described in Section 5.4, all other learners use the full training split. These methods produce only classification labels, so they enter the accuracy/F1 comparison of Section 6.1 and have no counterpart for the reliability metrics.

Recent LLM-based classifier reproductions. To position the perception layer against 2024–2025 LLM-based fault diagnosis classifiers, we reimplement two recent methods on this repository’s data splits: FD-LLM [8], a rank-4 LoRA fine-tune of Llama-3.1-8B-Instruct on 15 statistical signal features (learning rate 10^{-4} , bf16, 3 epochs), and T2MFDF [71], which fuses a time-series encoder with BERT-family fault-description embeddings. All reported values are our reproductions on repository splits over 3 seeds; no numbers are copied from the source papers. Both are classification-only methods without structured hypothesis–evidence outputs, so they are compared on accuracy and macro F1 (Section 6.1).

The perception layer shared by all pipeline arms is an XGBoost classifier over the hand-crafted features of Section 5.1, trained under the stratified cap of Section 5.4. Because every pipeline arm shares this layer, diagnostic accuracy within the paired comparison is set almost entirely by the perception stage (Table 7), and the experimental questions of this paper concern the reliability, action-quality, and governance metrics defined next; the standalone classification baselines above quantify what dedicated classifiers achieve on the same splits.

Scope of the comparison.. Beyond the two reproduced classifiers above, several recent LLM-based fault diagnosis systems are discussed in related work but not run as comparison arms, including UniFault [37], BearLLM [32], and

PHM-GPT [10]: they are single-domain or closed-source, and none exposes structured hypothesis–evidence outputs on which released-hallucination and action metrics are defined, which would reduce any comparison to the perception layer that all our arms already share.

5.3. Evaluation Metrics

We organize the evaluation metrics into four categories that collectively assess diagnostic capability and operational reliability. The preregistered comparison family comprises accuracy, macro F1, hallucination rate, action accuracy, and false-escalation rate.

Diagnostic performance. Top-1 accuracy (Acc) and macro-averaged F1 measure the correctness of the released fault identification.

Reliability and hallucination. The *hallucination rate* (Halluc.) is the fraction of all episodes in which a hypothesis whose stated evidence contradicts the actually retrieved data or the actual signal features is *released* downstream, that is, gated to proceed; deferred and escalated hypotheses never reach the operator, so verification can legitimately reduce this rate. The ungated per-hypothesis rate is reported alongside as the hypothesis-level hallucination rate and is identical across paired arms by construction. Hallucination is judged by an independent deterministic rule judge that never participates in the pipeline; Section 6.2 cross-checks it against an off-the-shelf NLI judge (deberta-v2-xlarge-mnli) and reports their agreement. The *contradiction rate* (Contra.) counts released episodes citing mutually inconsistent evidence, and the *overconfident error rate* (OvConf.) counts episodes with gating confidence $\alpha > 0.8$ but an incorrect final diagnosis.

Action quality and routing. *Action accuracy* (ActAcc) is the fraction of episodes whose recommended action level matches the ground-truth level; an escalated episode is credited as correct if and only if the ground-truth level is A3 or A4, so escalation is rewarded exactly when human involvement is warranted. Ground-truth action labels come from a deterministic rule table mapping each fault class and severity to a level A0–A4, derived from IEEE 1159 [23] severity guidance for PQ, ISO 10816 [67] vibration severity zones for CWRU, and IEC 61400 [68] alarm procedures for CARE; the table is implemented in code, disclosed in full in the supplementary material, and not fitted to any experimental data. The *false-escalation rate* (FalseEsc) is the fraction of all episodes escalated to human review although the ground-truth level was A0 or A1. Escalating a ground-truth A2 episode therefore counts as an incorrect action but not as a false escalation, since A2 sits between the autonomous and human-warranted bands. We additionally report the proceed, defer, and escalate rates that characterize the human-handoff cost.

LLMOps stability. The *regression pass rate* is the fraction of update cycles in which a proposed parameter update satisfies all acceptance tolerances on a target regression suite (Section 6.4); blocked updates correspond to rollbacks.

5.4. Implementation Details

The primary LLM backbone is Llama-3.1-8B-Instruct [72] (Hugging Face model ID: meta-llama/Llama-3.1-8B-Instruct), an instruction-tuned open-source model with 8 billion parameters, served locally via vLLM in FP16 precision. We select this model for three reasons. First, its open-source license enables full reproducibility and on-premise deployment without external API dependencies. Second, its strong instruction-following and structured-output capabilities are essential for generating JSON-formatted hypotheses. Third, its 8B parameter scale balances diagnostic capability with inference efficiency suitable for industrial deployment on commodity hardware. All LLM calls use temperature $T = 0.3$ for hypothesis generation and $T = 0.1$ for action planning; the self-consistency count M is part of the MOPSO decision vector and equals 7 at the deployed operating point. The maximum context length is set to 8,192 tokens.

To validate the framework’s backbone-agnostic design, we also evaluate CL-LLMOps with Mistral-7B-Instruct-v0.3 [70] as an alternative backbone. Mistral-7B uses sliding-window attention and grouped-query attention, a distinct architecture from Llama-3.1. All other pipeline components (CAEVN, FIS parameters, MOPSO-optimized thresholds, and the knowledge base) remain identical to the Llama-3.1 configuration, isolating the effect of the backbone swap. The multi-backbone results are reported in Table 8.

The neural verifier g_ϕ uses a cross-attention evidence verification architecture (CAEVN). The hypothesis and each evidence item are independently encoded by a frozen all-MiniLM-L6-v2 sentence encoder (384 dimensions). A 4-head cross-attention module ($d_k = 96$) computes an evidence-aware hypothesis representation, which a 2-layer MLP head (256 hidden units) maps through a sigmoid to a scalar support score in $[0, 1]$. A separate CAEVN is

trained per dataset on triples generated by running the backbone LLM over the training split with $M=1$: each of roughly 3,000 generated episodes contributes its retrieved-evidence triple together with a ground-truth-supporting and a confuser-contradicting evidence set, yielding a 9,000-triple pool per dataset (8,976 on PQ; 9,000 on CWRU and CARE). Episode labels are assigned by comparing the LLM’s stated diagnosis and cited evidence against the ground-truth fault label and the independent rule judge (PQ: 1,716 support / 1,276 contradict; CWRU: 2,209 / 790; CARE: 1,615 / 1,384; at most one neutral episode per dataset). Each checkpoint is trained for 50 epochs with AdamW (learning rate 10^{-3}) on a margin-ranking objective (margin 0.3) that scores supporting evidence above contradicting evidence for the same hypothesis, plus attention entropy regularization ($\beta = 0.01$, itself part of the MOPSO decision vector over a checkpoint grid) that discourages attention collapse onto a single evidence item. The CAEVN evaluates evidence–hypothesis consistency rather than diagnosis correctness, which prevents the verifier from merely replicating the LLM’s own confidence patterns. The MLP ablation verifier of Section 6.3 is trained on the same per-dataset pairs.

The fuzzy gating mechanism uses Mamdani inference with centroid defuzzification. Episodes whose semantic evidence coverage falls below the fixed coverage-defer threshold $\theta_p = 0.5$ are deferred as insufficient-evidence cases, and a severe-contradiction override escalates regardless of the defuzzified confidence (Eq. (9)). The six-dimensional decision vector $(\alpha_p, \alpha_d, \delta, \lambda_r, M, \beta)$ is optimized by multi-objective particle swarm optimization on the validation set of the PQ Classification dataset only; the membership-function breakpoints are expert-set and robustness-tested by simultaneous perturbation in Section 6.3. The MOPSO uses a swarm size of 50 and 100 iterations (5,050 pipeline evaluations per run), following the ≥ 5 particles-per-dimension guideline [20]. The deployed balanced solution, selected from the archive as the point closest to the normalized ideal, is $\alpha_p=0.50$, $\alpha_d=0.266$, $\delta=0$, $\lambda_r=0.1$, $M=7$, and $\beta=0.01$ (the continuous β request of 0.007 snaps to the nearest trained checkpoint grid value). This single operating point is applied uniformly across all three datasets without per-dataset re-tuning, mirroring deployment in which verification parameters are calibrated once and transferred to new equipment; Section 6.1 reports the resulting cross-domain transfer behavior, including its uneven handoff cost, and per-domain re-calibration is exercised by the regression experiment of Section 6.4.

The regression testing loop is evaluated with per-domain update cycles: a MOPSO re-tuning update derived on one domain’s validation split is replayed frozen against the regression suites of all three domains over 5 evaluation seeds, with acceptance tolerances of at most a 0.01 F1 drop, a 0.01 hallucination-rate increase, and a 0.02 false-escalation increase per suite (Section 6.4).

Statistical protocol. All headline comparisons follow a two-layer uncertainty design. The test-composition layer draws 10 class-stratified bootstrap evaluation partitions (seeds 100–109) of 160 episodes each from the untouched original test split, with the training and validation splits and all learned artifacts held fixed. The decoding layer repeats each evaluation under 3 decoding seeds, whose results are averaged within each partition before pairing. Paired arms share every alarm, sampled hypothesis set, and retrieved evidence item. Significance uses two-sided Wilcoxon signed-rank tests on the $N=10$ split-level paired differences with Holm step-down correction across the 15 dataset×metric comparisons of the preregistered family, and percentile-bootstrap 95% CIs on mean differences (10,000 resamples). With $N=10$ pairs and no ties, the smallest attainable two-sided raw p is 0.00195. A third layer probes the verifier’s training draw: five independent draws of the CAEVN training pool, each re-decoded from the training split and retrained from scratch, are evaluated on the paired comparison with the deployed operating point, the perception layer, and the evaluation episodes held fixed (Section 6.1). Component studies (ablation, verification baselines, backbone, and robustness) use the 3-decoding-seed protocol on 160-episode stratified evaluation subsets; the multi-objective optimizer comparison uses 10 paired runs. Each component study is an independent execution: within a study all arms share the evaluation subset and the LLM generations, so in-table comparisons are exactly paired, whereas batched vLLM decoding is not bitwise reproducible across executions. Across the five executions that report the full configuration (Table 6 and four component studies), its released hallucination varies by up to 0.015/0.008/0.040 on PQ/CWRU/CARE, so we draw conclusions only within a table.

All experiments are conducted on NVIDIA A800 80 GB GPUs. Feature extraction and signal processing use NumPy and SciPy; neural components are implemented in PyTorch 2.10, with the LLM served by vLLM 0.19 in FP16; sentence embeddings use `sentence-transformers`, and the knowledge base index uses CPU-resident FAISS. The shared XGBoost perception classifier is fitted on a class-stratified cap of 4,000 training windows per dataset. The RBF-kernel SVM alarm classifier included in the accompanying code is trained on a class-stratified cap of 20,000 samples, since the kernel matrix is quadratic in the training-set size and the full 168K PQ training split is computationally

Table 6: Paired comparison of Agent-noV and CL-LLMOps across three datasets. Values are mean $\pm$ s.d. over 10 stratified bootstrap test splits, each averaged over 3 decoding seeds (160 episodes per split). Acc: Top-1 accuracy (%); F1: macro F1 (%); Halluc.: released-hallucination rate; Contra.: released-contradiction rate; OvConf.: overconfident-error rate; ActAcc: action accuracy (%); FalseEsc: false-escalation rate; Proceed/Defer/Escal.: gating decision rates. Both arms share generations, so Acc/F1 are paired; Agent-noV releases every hypothesis (Proceed=1.0), which is also why its false-escalation rate is zero by construction rather than by merit. Lower released-hallucination, contradiction, and overconfident-error rates are in **bold**.

Dataset	Method	Acc	F1	Halluc.	Contra.	OvConf.	ActAcc	FalseEsc	Proceed	Defer	Escal.
PQ Class.	Agent-noV	93.0 $\pm$ 1.0	92.9 $\pm$ 1.1	0.437 $\pm$ 0.024	0.353 $\pm$ 0.021	0.070 $\pm$ 0.010	31.6 $\pm$ 1.0	0.000	1.000	0.000	0.000
	CL-LLMOps	93.0 $\pm$ 1.0	92.9 $\pm$ 1.1	0.177$\pm$0.019	0.161$\pm$0.016	0.012$\pm$0.004	30.4 $\pm$ 1.1	0.127 $\pm$ 0.003	0.418 $\pm$ 0.018	0.054 $\pm$ 0.005	0.529 $\pm$ 0.016
CWRU	Agent-noV	98.3 $\pm$ 0.7	98.3 $\pm$ 0.7	0.245 $\pm$ 0.011	0.192 $\pm$ 0.006	0.017 $\pm$ 0.007	41.0 $\pm$ 0.8	0.000	1.000	0.000	0.000
	CL-LLMOps	98.4 $\pm$ 0.6	98.4 $\pm$ 0.6	0.015$\pm$0.005	0.013$\pm$0.005	0.000$\pm$0.001	31.6 $\pm$ 0.4	0.389 $\pm$ 0.003	0.036 $\pm$ 0.007	0.032 $\pm$ 0.006	0.931 $\pm$ 0.009
CARE Wind	Agent-noV	57.0 $\pm$ 1.8	56.9 $\pm$ 1.7	0.509 $\pm$ 0.020	0.350 $\pm$ 0.019	0.430 $\pm$ 0.018	40.8 $\pm$ 1.9	0.000	1.000	0.000	0.000
	CL-LLMOps	57.5 $\pm$ 1.9	57.4 $\pm$ 1.8	0.100$\pm$0.015	0.062$\pm$0.012	0.002$\pm$0.002	15.1 $\pm$ 1.5	0.317 $\pm$ 0.012	0.155 $\pm$ 0.015	0.272 $\pm$ 0.016	0.573 $\pm$ 0.021

infeasible; linear-complexity learners are unaffected by this cap.

Knowledge base construction. The domain knowledge base $\mathcal{K}$ is indexed using FAISS with flat inner-product search over all-MiniLM-L6-v2 embeddings (384 dimensions). Each entry is a structured JSON document with a textual description, associated fault type tags, numerical feature ranges, and a source provenance field. We construct $\mathcal{K}$ through three complementary channels. The first channel collects curated fault signature descriptions from domain standards and textbooks: IEEE 1159 [23] disturbance characteristics for PQ, bearing defect frequency formulas from [7] for CWRU, and SCADA alarm interpretation guides from [30] for CARE. The second extracts case reports from the training split, each summarizing one labeled training window by its fault class and signal statistics (RMS, standard deviation, and peak). The third stores maintenance procedure templates that map fault types to recommended interventions with risk levels, sourced from ISO 10816 [67] for vibration and IEC 61400 [68] for wind turbines. Each fault class contributes one signature entry, three sampled case reports, and one procedure template, so the PQ, CWRU, and CARE knowledge bases contain 80, 50, and 10 entries respectively, deliberately compact bases representative of the curated documentation available at initial deployment. In production the knowledge base grows over time as new fault patterns are documented, and the FAISS index supports efficient online insertion without re-indexing. During evidence retrieval (Stage V1), the system queries $\mathcal{K}$ using the hypothesis text as the search query and returns the top- J entries by cosine similarity ($J=5$).

6. Results and Discussion

6.1. Main Results under the Paired Statistical Protocol

Table 6 reports the paired comparison between CL-LLMOps and the unverified agent baseline Agent-noV on all three datasets under the stratified-bootstrap protocol of Section 5.4. Because the two arms share every alarm, sampled hypothesis set, and retrieved evidence item, the comparison isolates exactly the contribution of the verification and gating layers: diagnostic accuracy and macro-F1 differ by at most 0.46 points (Table 7), and the arms differ mainly in which hypotheses are released, escalated, or deferred.

The headline finding is a large, statistically significant reduction of the released-hallucination rate on every dataset. On PQ Classification the rate falls from 0.437 to 0.177, on CWRU from 0.245 to 0.015, and on CARE from 0.509 to 0.100; the paired reductions of 26.0, 23.1, and 40.9 percentage points are all significant at the Holm-corrected floor $p_{\text{Holm}}=0.029$ of the $N=10$ design (Table 7). The hypothesis-level hallucination rate before gating is identical in both arms by construction (0.437, 0.245, and 0.509), so the entire reduction is attributable to the verification and gating layers withholding unsupported hypotheses from release. Overconfident errors show the same pattern: on CARE, where the noisy SCADA signals make the unverified agent wrong-but-confident on 43.0% of episodes, gating reduces this failure mode to 0.2%. These intervals quantify two sources of variability and no others: evaluation-set composition, through the stratified bootstrap partitions, and decoding stochasticity, through the three seeds averaged per partition. The training and validation splits and every learned artifact stay fixed across partitions, because each partition costs a full LLM pass over both arms. We therefore probe the training draw of the component under test separately: five independent draws of the CAEVN training pool, each re-decoded and retrained from scratch and

Table 7: Paired differences (CL-LLMOps minus Agent-noV, in percentage points) with percentile-bootstrap 95% CIs on the mean difference (10,000 resamples) and two-sided Wilcoxon signed-rank p -values, Holm-corrected across the 15-comparison family. With $N=10$ split pairs and no ties, the smallest attainable raw p is 0.00195; the corresponding Holm-adjusted floor is 0.029.

Metric	PQ Classification			CWRU Bearing			CARE Wind		
	Δ	95% CI	p_{Holm}	Δ	95% CI	p_{Holm}	Δ	95% CI	p_{Holm}
Accuracy	0.00	[0.00, 0.00]	1.000	+0.06	[0.00, +0.17]	1.000	+0.46	[+0.04, +0.88]	0.451
Macro F1	0.00	[0.00, 0.00]	1.000	+0.06	[0.00, +0.17]	0.500	+0.45	[+0.05, +0.89]	0.451
Hallucination rate	-25.96	[-26.58, -25.29]	0.029	-23.06	[-23.58, -22.52]	0.029	-40.90	[-41.48, -40.15]	0.029
Action accuracy	-1.27	[-2.27, -0.25]	0.451	-9.38	[-9.83, -8.90]	0.029	-25.71	[-26.90, -24.46]	0.029
False escalation	+12.69	[+12.54, +12.85]	0.029	+38.92	[+38.75, +39.08]	0.029	+31.73	[+31.02, +32.42]	0.029

evaluated with the deployed operating point held fixed, give released-hallucination reductions of 25.5 ± 2.7 , 24.3 ± 1.5 , and 42.7 ± 1.3 percentage points on PQ, CWRU, and CARE, with the deployed verifier’s reductions inside the five-fold range on PQ and CWRU and 0.35 points below it on CARE. What no layer of this design resamples is the perception classifier and the LLM generations, which would require re-decoding the corpus once per fold.

This safety gain has an explicit and quantified cost. CL-LLMOps hands off 58.3% of PQ episodes to a human (52.9% escalated, 5.4% deferred), and its false-escalation rate, the fraction of episodes escalated although the ground-truth action level was A0 or A1, is 0.127 on PQ, 0.389 on CWRU, and 0.317 on CARE. Action accuracy consequently drops relative to the release-everything baseline on CWRU (-9.4 points) and CARE (-25.7 points), because escalation is credited as correct only when the ground-truth action level warrants human involvement. We report this trade-off directly rather than netting it out: the framework buys hallucination containment with human workload, and the operating point on that trade-off curve is an explicit optimization target (Section 6.3).

Uneven cross-dataset transfer of the operating point.. A second finding we report plainly is that the MOPSO operating point, tuned once on the PQ validation set and transferred unchanged, behaves unevenly across domains. On PQ it proceeds autonomously on 41.8% of episodes; on CARE the proceed rate drops to 15.5%, and on CWRU to 3.6%, with 93.1% of CWRU episodes escalated. The transferred gate is therefore strongly over-conservative outside its tuning domain: on CWRU it achieves a near-zero released-hallucination rate (0.015) mostly by declining to release. The obvious remedy is per-domain re-calibration, and the cross-domain regression protocol of Section 6.4 already measures it on each domain’s own regression suite, whose composition differs from the test splits of Table 6. It does not work. Re-tuning on a domain’s own validation split under the same objectives and knee-selection rule makes the gate more conservative still: the proceed rate moves from 0.380 to 0.045 on PQ and from 0.027 to 0.007 on CWRU, and on CARE it stays put while action accuracy falls from 14.2% to 3.8%. The conservatism therefore follows from pricing every handoff in the objective and selecting the knee of the front, not from transferring a foreign operating point, and recovering autonomy calls for a different selection rule or objective weighting rather than local tuning. We regard the uneven transfer as a limitation of the single-operating-point deployment mode rather than of the verification architecture itself.

Multi-backbone evaluation.. To check the model-agnostic design, we replicate the pipeline with Mistral-7B-Instruct-v0.3 [70] as an alternative backbone under the 3-seed component protocol, without re-tuning CAEVN, FIS, or MOPSO parameters. Diagnostic accuracy is unchanged by construction, since the shared perception classifier is backbone-independent. As shown in Table 8, the verification effect transfers: released hallucination stays far below the unverified rates of Table 6 on all three datasets. The absolute rates differ across backbones because the generated hypotheses differ; Mistral-7B produces fewer rule-judge-flagged hypotheses on PQ and CARE and more on CWRU, and its gate routes more CARE episodes to defer (0.381) than the Llama-3.1 configuration.

Comparison with standalone classification baselines.. Table 9 places the shared perception layer against the classical, deep, and reproduced 2024–2025 LLM-based classifiers of Section 5.2. On the two multi-class benchmarks, dedicated classifiers trained on the full split are strong: the 1D-CNN reaches 99.1% on PQ and saturates CWRU at 100%, the Transformer reaches 99.6% on PQ, and full-split XGBoost reaches 98.2%/99.0%. The pipeline’s perception layer is the same XGBoost learner deliberately trained under the 4,000-window stratified cap of Section 5.4, capped to

Table 8: Multi-backbone evaluation of CL-LLMOps (mean of 3 decoding seeds, 160-episode stratified evaluation subsets). All components except the LLM backbone are identical; accuracy is set by the shared perception classifier. Halluc.: released-hallucination rate; ActAcc: action accuracy (%); FalseEsc: false-escalation rate.

Dataset	Backbone	Acc	Halluc.	ActAcc	FalseEsc	Proceed
PQ Class.	Llama-3.1-8B	93.3	0.165	31.9	0.127	0.406
	Mistral-7B	93.3	0.085	33.1	0.094	0.379
CWRU	Llama-3.1-8B	98.5	0.017	31.9	0.394	0.038
	Mistral-7B	98.5	0.013	32.9	0.373	0.027
CARE Wind	Llama-3.1-8B	59.0	0.073	14.8	0.331	0.127
	Mistral-7B	59.0	0.027	24.6	0.231	0.215

Table 9: Standalone classification baselines vs. the shared perception layer (%). Top block: classical and deep baselines, single deterministic run (seed 42) on the full held-out test splits (36,001/883/34,746 samples); SVM trained under the 20,000-sample stratified cap, all others on the full training split. Middle block: reproduced LLM-based classifiers (mean $\pm$ s.d., 3 seeds) on repository splits; no numbers copied from the source papers; FD-LLM is fine-tuned on subsampled training sets (3,200/2,000/400) and evaluated on 800/500/100 test samples, T2MFDF on 3,200/835/400 test samples. Bottom: the capped perception layer under the bootstrap protocol of Table 6. Evaluation sets differ across blocks, so per-column bolding is omitted.

Method	PQ Class.		CWRU		CARE Wind	
	Acc	F1	Acc	F1	Acc	F1
SVM (RBF, 20K cap)	96.5	96.4	96.4	95.5	58.8	56.6
XGBoost (full split)	98.2	98.2	99.0	98.7	56.6	54.5
ANFIS [43]	77.5	76.8	84.6	81.6	60.2	56.1
1D-CNN	99.1	99.1	100.0	100.0	47.4	47.3
Transformer	99.6	99.6	96.9	96.2	61.1	37.9
FD-LLM (repro.) [8]	64.9 $\pm$ 1.7	63.8 $\pm$ 1.9	90.7 $\pm$ 0.9	90.5 $\pm$ 1.0	50.7 $\pm$ 3.1	49.0 $\pm$ 5.1
T2MFDF (repro.) [71]	92.4 $\pm$ 2.1	92.1 $\pm$ 2.4	99.96 $\pm$ 0.07	99.95 $\pm$ 0.08	57.6 $\pm$ 2.9	56.7 $\pm$ 3.5
Perception layer (ours, 4K cap)	93.0 $\pm$ 1.0	92.9 $\pm$ 1.1	98.4 $\pm$ 0.6	98.4 $\pm$ 0.6	57.5 $\pm$ 1.9	57.4 $\pm$ 1.8

keep the repeated full-pipeline campaigns affordable, which costs about five points on PQ (93.0% vs. 98.2%); we disclose this gap rather than tuning it away, since the pipeline experiments hold the perception layer fixed across all arms. On the noisy CARE task every standalone classifier degrades or collapses: the Transformer’s 61.1% accuracy with 37.9 F1 is consistent with near-majority-class prediction (the test split is 61.1% negative), the 1D-CNN falls to 47.4%, and no method exceeds 60.2% accuracy with a balanced F1. Among the LLM-based reproductions, T2MFDF is the strongest overall (92.4%/99.96%/57.6%), exceeding the capped perception layer on CWRU, while the FD-LLM LoRA reproduction trails on all three datasets under its subsampled fine-tuning budget. Two readings follow. First, classification strength on these benchmarks is attainable by several routes, and a stronger classifier such as the 1D-CNN or T2MFDF could directly replace the perception layer inside CL-LLMOps. Second, none of these methods produces the structured hypothesis–evidence outputs on which released-hallucination, escalation, and action metrics are defined, so the reliability results of Tables 6–7, which are the subject of this paper, have no counterpart in these systems.

6.2. Reliability Analysis and Verification-Layer Baselines

Independent hallucination judging. The hallucination metrics in Table 6 are produced by a deterministic rule-based judge that is fully separate from the deployed CAEVN/FIS verifier: it checks each released hypothesis’s cited evidence against the actually retrieved knowledge-base items and the actual signal feature values. As an independent cross-check we re-judge the hypotheses of all 1,440 audited episodes, released or withheld, with an off-the-shelf NLI model (deberta-v2-xlarge-mnli) that likewise never participates in the pipeline. The two automated judges agree on 54.8% of verdicts (Cohen’s $\kappa=0.11$; per-dataset raw agreement 0.60/0.46/0.59 on PQ/CWRU/CARE), with the rule judge flagging 37.2% of episodes versus 53.6% for the NLI judge. The low chance-corrected agreement shows that automated hallucination judging is itself measurement-sensitive; we therefore report the conservative rule judge as the headline metric and release a 120-episode disagreement-oversampled audit packet with the code so that human

Table 10: Verification-layer baselines (mean of 3 decoding seeds). All arms share the perception layer and LLM generations and differ only in the verification/gating back-end. Halluc.: released-hallucination rate; FalseEsc: false-escalation rate; ActAcc: action accuracy (%). The NLI-verifier gate escalates every episode on all three datasets, so its zero hallucination is degenerate (Proceed=0).

Gate	PQ Classification				CWRU Bearing				CARE Wind			
	Halluc.	FalseEsc	Proceed	ActAcc	Halluc.	FalseEsc	Proceed	ActAcc	Halluc.	FalseEsc	Proceed	ActAcc
FIS gate (ours)	0.163	0.133	0.423	28.8	0.021	0.388	0.052	32.5	0.113	0.321	0.171	16.5
Calibrated (Platt)	0.154	0.102	0.400	33.8	0.008	0.383	0.029	26.9	0.067	0.340	0.148	14.6
Calibrated (isotonic)	0.277	0.056	0.777	31.9	0.083	0.317	0.329	33.5	0.408	0.063	0.848	40.4
Guardrail filter	0.425	0.000	1.000	33.8	0.263	0.008	0.990	42.3	0.494	0.000	1.000	46.3
NLI-verifier gate	0.000	0.188	0.000	43.8	0.000	0.400	0.000	30.0	0.000	0.500	0.000	0.0

adjudication can be carried out independently. We report no human agreement statistics of our own and do not substitute automated agreement for human agreement. What we can settle without human labels is whether the reported effect depends on the judge: re-deriving both arms on the same episodes and the same gate decisions under each judge in turn leaves the finding intact on every dataset, with a released-hallucination reduction of 22.9 versus 21.5 percentage points under the rule and NLI judges on PQ, 20.6 versus 60.8 on CWRU, and 39.2 versus 56.5 on CARE. The judges disagree about how many hypotheses are unsupported, not about whether gating withholds them, and the judge we report is the conservative choice on two of the three datasets.

Comparison with verification-layer alternatives. Table 10 compares the FIS gate against the alternative verification back-ends of Section 5.2 under the same 3-seed protocol: calibrated probability gating (Platt and isotonic post-processing of the same verification scores, with the operating point coverage-matched on validation), a static safety-guardrail filter, and an off-the-shelf NLI entailment verifier used as the gate.

Three observations follow. First, the static guardrail filter provides essentially no hallucination containment (released rates 0.425/0.263/0.494, close to the unverified agent), confirming that keyword-level safety filtering does not address evidence fabrication. Second, the off-the-shelf NLI gate collapses to escalate-everything on all three datasets: its zero released hallucination is achieved at proceed rate zero, the degenerate corner that the handoff-priced MOPSO objective is designed to exclude. Third, calibrated Platt gating over the same verification scores is a genuinely competitive baseline at matched coverage: it attains lower released hallucination than the FIS on all three datasets (0.154 vs. 0.163 on PQ, 0.008 vs. 0.021 on CWRU, and 0.067 vs. 0.113 on CARE) while proceeding on about 2.3 points fewer episodes. The PQ margin is within the cross-study variation of the full configuration reported in Section 5.4; the CWRU and CARE margins exceed it. We therefore read the outcome as parity on PQ and a Platt advantage on CWRU and CARE. Isotonic calibration is unstable in the opposite direction, drifting to high-coverage operating points with 1.7 to 3.6 times higher released hallucination than the FIS on PQ and CARE. The empirical case for the fuzzy gate therefore does not rest on dominating well-calibrated scalar gating on these aggregate metrics; it rests on the auditable rule semantics, the first-class three-way decision, and the stability arguments of Section 4.3.3, although its aggregate safety performance trails the best calibrated alternative on CWRU and CARE.

6.3. Ablation Study

Table 11 reports the component ablation across all three datasets under the 3-seed protocol. Because all variants share the perception layer and LLM generations, accuracy and macro F1 are nearly variant-invariant (within ± 0.2 , ± 0.0 , and ± 1.3 points of the full model on PQ/CWRU/CARE); the informative movement is in the safety and routing metrics.

Removing the verifier entirely reverts every dataset to the release-everything regime, with released hallucination rising from 0.165 to 0.404 on PQ, from 0.017 to 0.256 on CWRU, and from 0.073 to 0.506 on CARE. Removing only the fuzzy gate while keeping verification scores raises PQ hallucination to 0.283, and dropping the CAEVN layer while keeping rule checks (rule-only) raises it to 0.300 on PQ and 0.360 on CARE; the model-only variant degrades more mildly, indicating that the learned layer carries most of the filtering signal while the rule layer contributes complementary precision. Replacing CAEVN with the mean-pooling MLP increases PQ hallucination to 0.190 and destabilizes CARE routing, where false escalation climbs to 0.435 and action accuracy collapses to 4.8%. Hard

Table 11: Ablation study (mean of 3 decoding seeds). Each row removes or replaces one component of the full CL-LLMOps. “Rule-only” and “model-only” keep a single verification layer; “MLP verifier” replaces CAEVN with a mean-pooling MLP; “hard gating” replaces the FIS with fixed thresholds; “grid search” replaces the MOPSO operating point with the best coarse-grid corner. Halluc.: released-hallucination rate; FalseEsc: false-escalation rate; ActAcc: action accuracy (%). Variants sit at different points of the safety–autonomy trade-off, so the lowest hallucination in a column is not the best arm and no value is bolded.

Variant	PQ Classification				CWRU Bearing				CARE Wind			
	Halluc.	FalseEsc	Proceed	ActAcc	Halluc.	FalseEsc	Proceed	ActAcc	Halluc.	FalseEsc	Proceed	ActAcc
Full CL-LLMOps	0.165	0.127	0.406	31.9	0.017	0.394	0.038	31.9	0.073	0.331	0.127	14.8
w/o Verifier	0.404	0.000	1.000	32.7	0.256	0.000	1.000	42.1	0.506	0.000	1.000	43.1
w/o Gating	0.283	0.094	0.615	30.2	0.110	0.265	0.485	37.1	0.148	0.327	0.363	12.3
w/o Self-consistency	0.169	0.125	0.394	31.0	0.023	0.377	0.046	33.5	0.100	0.346	0.146	15.6
w/o Regression testing	0.163	0.125	0.448	29.4	0.017	0.388	0.048	32.1	0.098	0.323	0.158	13.8
Rule-only verifier	0.300	0.035	0.663	31.3	0.085	0.008	0.413	39.6	0.360	0.192	0.606	27.1
Model-only verifier	0.177	0.123	0.433	31.0	0.015	0.388	0.052	32.1	0.085	0.323	0.146	12.7
MLP verifier (repl. CAEVN)	0.190	0.156	0.477	35.4	0.033	0.310	0.117	28.5	0.056	0.435	0.075	4.8
Hard gating (repl. FIS)	0.215	0.123	0.492	31.9	0.025	0.383	0.085	33.5	0.123	0.419	0.202	4.8
Grid search (repl. MOPSO)	0.048	0.181	0.140	38.8	0.013	0.392	0.019	31.3	0.002	0.388	0.006	11.3

thresholds in place of the FIS raise PQ hallucination to 0.215 and CARE false escalation to 0.419. Removing self-consistency or regression testing moves the safety metrics only marginally under this protocol; self-consistency is an adopted technique whose contribution here is modest, and the value of regression testing is longitudinal (Section 6.4) rather than per-episode.

The grid-search row deserves an honest reading: the best coarse-grid corner is not dominated by the MOPSO knee on the per-dataset safety metrics. It reaches lower released hallucination (0.048 on PQ) by moving to a far more conservative operating point, proceeding on only 14.0% of PQ episodes and 0.6% of CARE episodes with higher false escalation (0.181 on PQ). It is a different point on the same safety–autonomy trade-off rather than a better optimizer outcome; the value of MOPSO is that it exposes the whole front of such points and quantifies their cost, which a single grid corner cannot do (Section 6.3).

Robustness to non-Ruspini partition perturbation. The fuzzy partition is deliberately non-Ruspini (Section 4.3.3). Shifting all six score/coverage membership breakpoints simultaneously by ± 0.05 and ± 0.10 changes released hallucination by at most 0.038 (PQ), 0.004 (CWRU), and 0.023 (CARE) relative to the unshifted gate, with accuracy unchanged on PQ/CWRU and within 0.2 points on CARE, and proceed rates moving by at most 0.094. The partition diagnostics confirm the intended geometry: membership sums exceed 1 nowhere, and fall below 1 on 49.8% of the input range, which is the designed ambiguity-to-escalation zone.

MOPSO optimization analysis. Fig. 5 shows the Pareto front from the production MOPSO run on the PQ validation set. The archive converges to 8 unique objective vectors. Classification F_1 is constant at 0.935 across the entire front, because the released diagnosis labels come from the shared perception layer; the optimization therefore resolves into a two-dimensional trade-off between released hallucination and human handoff. Along the front, hallucination falls from 0.179 to 0 as the handoff rate rises from 0.545 to 1.0. The selected balanced solution, closest to the normalized ideal point, sits at the knee with handoff 0.589 and hallucination 0.107 ($\alpha_p=0.50$, $\alpha_d=0.266$, $\delta=0$, $\lambda_r=0.1$, $M=7$, $\beta=0.01$). The high-risk margin is active on several non-knee front solutions, with δ ranging from 0.02 to 0.14, confirming that the severity-conditioned gate shift participates in shaping the conservative end of the front.

Table 12 compares MOPSO against NSGA-II over 10 paired runs with an identical evaluation budget of 5,050 pipeline evaluations per run, using the combined non-dominated union of all runs as the reference front. MOPSO attains significantly better hypervolume (0.820 vs. 0.593) and IGD (0.067 vs. 0.187), both at the Holm-corrected $p=0.006$ floor of the 10-run Wilcoxon design. NSGA-II records the lower generalized spread, 1.000 against 1.404, but that number is the degenerate value rather than an advantage: Δ^* evaluates to exactly 1 whenever every point of a front has a duplicate, and NSGA-II returns exactly 1.000 with zero variance in all ten runs. MOPSO instead keeps 12 to 30 non-dominated archive entries per run, from which an operating point is selected; the archive is not de-duplicated, and the stored archive of the production run holds 8 distinct objective vectors. We report the spread

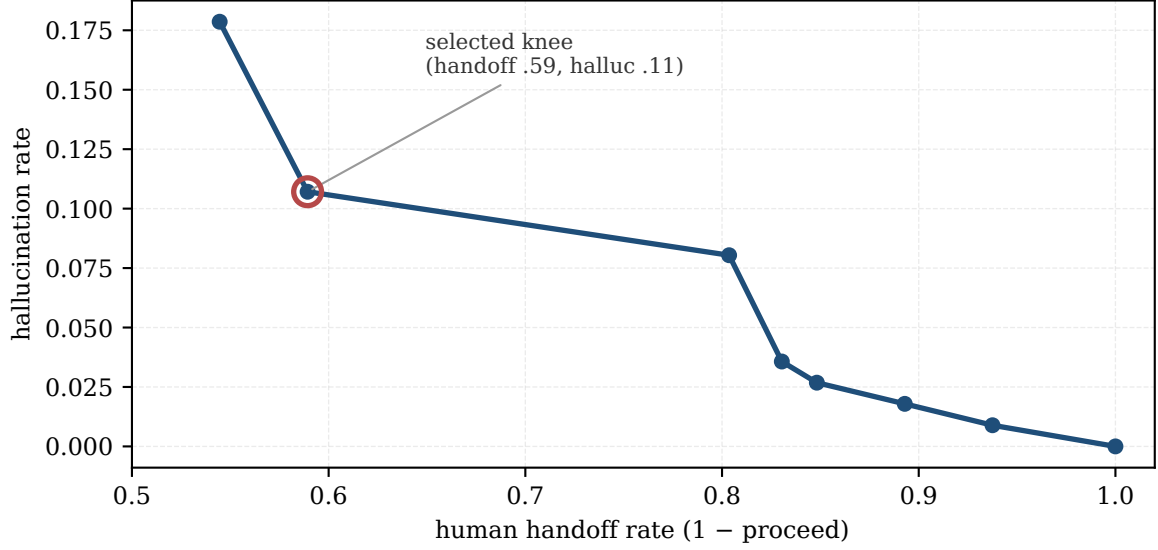


Figure 5: MOPSO Pareto front on the PQ validation set (production run, seed 42; 8 unique non-dominated objective vectors). Classification F_1 is constant at 0.935 across the front, so the three-objective problem of Eq. (11) reduces to the plotted two-dimensional trade-off between released hallucination and human handoff. The circled knee solution (handoff 0.589, hallucination 0.107) is the deployed operating point.

Table 12: Multi-objective optimizer comparison on the PQ validation set: mean $\pm$ s.d. over 10 paired runs at a matched budget of 5,050 evaluations per run. HV: hypervolume (higher is better); IGD: inverted generational distance (lower is better); Spread: generalized Δ^* (lower is better). All three differences reach the Holm-corrected Wilcoxon floor $p=0.006$, but the NSGA-II spread of exactly 1.000 in every run is the degenerate value of Δ^* for a front of duplicated points, not an even distribution, so only the hypervolume and IGD advantages are bolded.

Optimizer	HV $\uparrow$	IGD $\downarrow$	Spread $\downarrow$
MOPSO (ours)	0.820 ± 0.034	0.067 ± 0.019	1.404 ± 0.214
NSGA-II	0.593 ± 0.008	0.187 ± 0.008	1.000 ± 0.000

column for completeness and draw no diversity conclusion from it. Fig. 6 shows the per-run distributions behind the means: the hypervolume and IGD gaps separate the two optimizers with no overlap across the ten runs.

6.4. LLMOps Effectiveness

Computational cost. End-to-end pipeline latency under the statistical protocol averages 8.3 ± 0.6 s per episode on PQ, 6.7 ± 0.2 s on CWRU, and 7.9 ± 0.5 s on CARE, dominated by the $M=7$ self-consistency hypothesis decodes plus the action and audit-explanation calls on a single A800 GPU serving Llama-3.1-8B via vLLM. These latencies are acceptable for post-event diagnostic analysis, where decision timescales range from minutes to hours, and batch decoding across concurrent alarms amortizes the cost further.

Cross-domain regression gating. We evaluate the Stage-R regression gate with per-domain update cycles: for each source domain, a MOPSO re-tuning update is derived on that domain’s validation split and then replayed frozen against the regression suites of all three domains over 5 seeds, with acceptance tolerances of at most a 0.01 F1 drop, a 0.01 hallucination increase, and a 0.02 false-escalation increase. Table 13 shows the resulting pass-rate matrix.

The matrix is deliberately strict and mostly red, and this is the intended behavior of a safety gate rather than a failure of the updates. The PQ-sourced update, for example, cuts released hallucination on its own suite by 13.0 points on average, but raises false escalation by 6.1 points, exceeding the 0.02 tolerance, so the gate blocks deployment; the CARE-sourced update raises CARE false escalation by 10.3 points and is likewise blocked. Only the CWRU suite, where the transferred gate already operates at very low proceed rates, tolerates all updates. The experiment demonstrates the core LLMOps claim in measured form: single-domain re-tuning routinely produces side effects on the

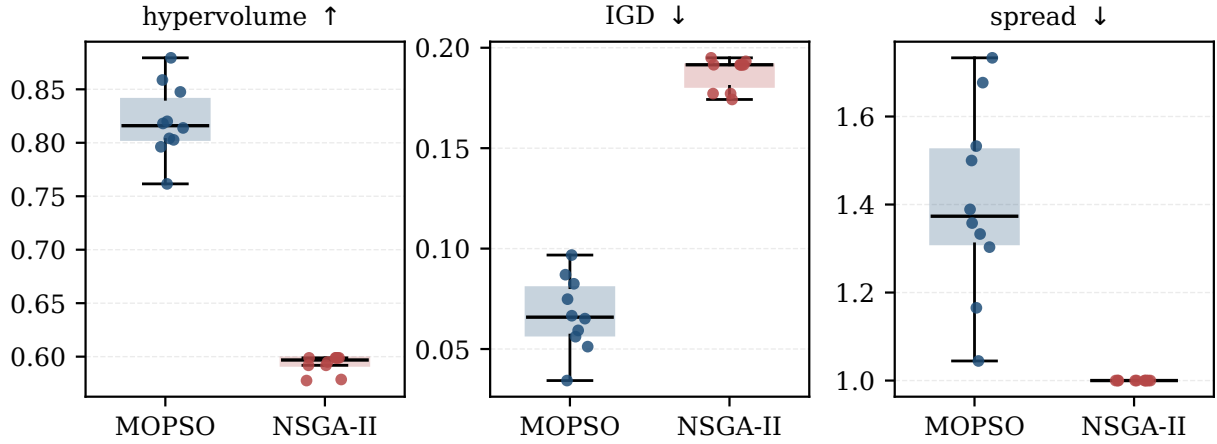


Figure 6: Per-run indicator distributions over the 10 paired optimizer runs behind Table 12, with every run drawn as a point beside its box. All three differences reach the Holm-corrected floor of the design (Table 12). Hypervolume and IGD separate the optimizers without overlap. The NSGA-II spread column collapses to exactly 1.000 in all ten runs, the value Δ^* takes when a front consists of duplicated points.

Table 13: Cross-domain regression gate pass rates (5 update cycles per cell). Rows: domain whose validation split sourced the parameter update; columns: target regression suite. A cell passes only if F1, hallucination, and false-escalation tolerances all hold on that suite.

Update source	Target suite		
	PQ	CWRU	CARE
PQ	0.0	1.0	0.0
CWRU	0.0	1.0	0.0
CARE	0.2	1.0	0.0

safety metrics that per-domain accuracy tracking would miss, and the regression gate detects and blocks exactly these cases before deployment. It also quantifies the earlier transfer finding: outside the CWRU suite, whose transferred gate already proceeds rarely, no cross-domain update passes in more than one of five cycles.

7. Conclusion and Future Work

We presented CL-LLMOps, a closed-loop LLMOps framework that couples LLM-powered diagnosis with the operational dependability required for industrial deployment. Unlike prior approaches that stop at classification or one-shot recommendation, our framework coordinates the full diagnostic lifecycle across five stages: alarm normalization, hypothesis generation, evidence-driven verification, action planning, and regression testing, with test sets that evolve automatically across cycles. The central technical contribution is a hierarchical verification mechanism that combines rule-based checking, a cross-attention evidence verifier, and LLM-assisted explanation. A Mamdani fuzzy inference gate unifies these layers by mapping continuous action confidence scores to proceed, escalate, or defer decisions, and the full parameter space is jointly tuned through multi-objective particle swarm optimization. This design keeps the LLM in a supportive role rather than an authoritative one, which prevents hallucinated reasoning from overriding quantitative evidence assessments.

Experimental validation covers three industrial domains: power quality disturbance classification, bearing fault diagnosis, and wind turbine SCADA anomaly detection. Under a paired protocol in which the verified and unverified arms share all generations, verification cuts the released-hallucination rate from 43.7% to 17.7% on the primary power quality benchmark, from 24.5% to 1.5% on CWRU, and from 50.9% to 10.0% on CARE, all significant under Holm-corrected Wilcoxon tests, and reduces overconfident errors on the noisiest SCADA task from 43.0% to 0.2% of episodes. These gains carry an explicit human-handoff cost, ranging from 58% of episodes on the tuning domain to 96% under cross-domain transfer of the operating point, which we report as a limitation of single-point deployment

and expose as an optimization target through the Pareto front. Ablation studies show that the learned verifier carries most of the filtering signal with the rule layer adding complementary precision, that removing gating or verification raises the released-hallucination rate by factors ranging from roughly two on the tuning domain to more than an order of magnitude on the bearing task, and that calibrated probability gating is a competitive scalar alternative whose aggregate safety matches the fuzzy gate on PQ and exceeds it on CWRU and CARE; the case for the fuzzy formulation rests on its auditable rule semantics and first-class three-way decision. The cross-domain regression gate blocks measured parameter updates whose false-escalation side effects would escape accuracy-only tracking, and extension from standard-conformant waveforms to field recordings is a natural next step supported by recent cross-platform robustness studies [2]. The modular architecture supports incremental adoption: the verification layer plugs into an existing LLM pipeline ahead of the full closed-loop governance, and integration with commercial PHM systems is a planned extension. The CAEVN training set is generated by the Llama-3.1-8B backbone used at inference time, so transfer to other backbones uses a held-out hypothesis pool or a brief recalibration pass, with Table 8 providing initial cross-backbone evidence.

From an Industry 5.0 perspective, the framework advances three pillars. *Human-centricity*: the escalation interface routes uncertain diagnoses to human experts with interpretable fuzzy confidence scores and attention-highlighted evidence rather than opaque classification outputs. *Resilience*: the regression testing loop detects and rolls back degradations before they propagate to operations. *Sustainability*: the graded gating strategy prevents over-maintenance triggered by false-positive diagnoses, cutting unnecessary interventions. End-to-end latency averages below 9 s per episode, which is small against the minutes-to-hours decision timescales of post-event diagnostic analysis (Section 6.4).

Several directions remain for future work. First, incorporating physics-based simulation models with interval Type-2 fuzzy sets [46] for the gating layer to handle multi-site deployment uncertainty. Second, human adjudication of the released audit packet, together with active learning that feeds expert verdicts back into the verifier, which aligns with the human-centric vision of Industry 5.0 [16]. Third, extending multi-backbone coverage beyond Llama-3.1-8B and Mistral-7B to commercial backbones such as GPT-4 [73], since the CAEVN and FIS parameters transfer without re-tuning. Fourth, developing a knowledge base maintenance protocol with human-validated quality gates for new entries and periodic verifier recalibration to address concept drift as equipment ages and fault signatures evolve over multi-year deployments. Fifth, extending the regression gate beyond the parameter updates evaluated here to prompt revisions and retrieval-index changes, and over longitudinal horizons in which the suite itself grows across many update cycles. Sixth, advancing toward functional safety certification under standards such as IEC 61508: the regression testing and version governance mechanisms in Stage R provide a foundation for the systematic validation that LLM-based diagnostic systems require.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. For full transparency, S. Li is employed by China Southern Power Grid Company Limited, which funded this study through the Science and Technology Project ZBKJXM20220010 as acknowledged above; this affiliation and funding relationship is fully disclosed and did not influence the experimental design, data analysis, or reporting of results.

Data Availability

The CWRU and CARE datasets are publicly available; the PQ dataset is synthetic. The Power Quality (PQ) dataset was synthesized following IEEE Std 1159-2019 [23] specifications; the synthesis scripts with all parametric model coefficients and noise configurations are available at <https://github.com/qyjjx/CL-LLM0ps>. The CWRU

Bearing dataset is available from the Case Western Reserve University Bearing Data Center [6]. The CARE Wind Turbine SCADA dataset is available from its Zenodo deposit [29]; this work uses release v6 (record 15846963).

Code Availability

The complete source code for CL-LLMOps, including the CAEVN training pipeline, fuzzy gating implementation, MOPSO optimizer, regression testing framework, prompt templates, and all baseline reproduction scripts, is available at <https://github.com/qyijx/CL-LLMOps>.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used Claude (Anthropic) for language polishing of selected passages, cross-checking of numerical consistency between the manuscript and result files, and code-level edits limited to comment cleanup and variable renaming in pipeline scripts. The AI tool was not involved in experimental design, data analysis, result generation, or the formulation of scientific claims. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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